Macroprudential policies and homeownership rates

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Abstract

Macroprudential policies are a key tool of financial regulators, but concerns persist that they restrict access to homeownership. I examine this concern for 28 European countries over 2003 to 2017, using a narrative index of macroprudential actions and event studies around five large borrower-based reforms. Recorded actions are not followed by detectable changes in homeownership rates, in aggregate or for young, mortgaged or low-income households, and the estimates rule out cumulative declines of more than half a percentage point over four years. The aggregate rate does not fall after the large reforms either, and neither actions nor reforms leave a common trace in aggregate credit or house prices, although the micro literature documents large effects on targeted borrowers. Borrower-level effects need not aggregate, because lenders reallocate credit, excluded buyers enter later, prices absorb part of the demand shift, and the homeownership stock dilutes the flow of first-time buyers. The results do not support concerns that macroprudential policies have reduced homeownership in aggregate, while leaving open effects on who obtains credit and on the timing of entry.

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1 Introduction

Recorded changes in macroprudential regulation are not followed by detectable changes in homeownership rates. This paper documents that result for 28 European countries over 2003–2017, and shows that the estimates are precise enough to rule out aggregate effects of the size predicted by a widely used class of structural housing models. The question matters because borrower-based measures such as loan-to-value and debt-service-to-income limits have become standard tools of financial regulation, and because they are widely believed to come at a cost. Macroprudential policies appear successful in limiting credit growth and house price growth (Arena *et al.* , 2020), but the concern is that they exclude first-time buyers and low-income households from homeownership. Several countries have relaxed loan-to-value limits or introduced exemptions for first-time buyers on these grounds. Yet the empirical literature on macroprudential policy focuses almost exclusively on credit and house prices. Araujo *et al.* (2024) collect 6,000 estimates from 58 studies of the effects of macroprudential policies and do not mention homeownership once.[1]

I combine the Macroprudential Policies Evaluation Database (MaPPED) of the European Central Bank (Budnik & Kleibl, 2018) with homeownership data from the European Union Statistics on Income and Living Conditions (EU-SILC). Following the narrative approach of Fernández-Gallardo *et al.* (2025), I build a country-level index from policy actions that were not taken with a counter-cyclical motive, and estimate local projections of the cumulative change in the homeownership rate on changes in this index over horizons of one to five years. The sample of 28 countries is larger than in most cross-country studies of macroprudential policy, and the with-controls specifications retain all 28 countries. The cumulative response of the aggregate

[1] A large number of studies have examined the impact of macroprudential policies on credit and house price growth (see Galati & Moessner, 2013, 2018; Araujo *et al.* , 2024, for an overview). Kuttner & Shim (2016) find significant negative effects of policy tightening on housing credit and house price growth. Lim *et al.* (2011) show that various macroprudential tools affect the cyclicality of private sector credit and leverage. Alam *et al.* (2019) demonstrate that tighter Loan-to-Value (LTV) constraints have non-linear effects on credit. Igan & Kang (2011) find that transactions and house prices drop significantly after debt-to-income (DTI) and LTV tightening. Other studies include Cerutti *et al.* (2017), Akinci & Olmstead-Rumsey (2018), and Poghosyan (2020). Tzur-Ilan (2023) show that higher LTV limits resulted in borrowers choosing more affordable housing units. Laufer & Tzur-Ilan (2021) find that higher risk-weights on high-LTV mortgages slow house price growth. Peydró *et al.* (2024) show that caps on high loan-to-income mortgages reduced credit supply and house price growth. To my knowledge, none of these studies have linked macroprudential policies to homeownership rates.

homeownership rate to a one-unit change in the index is between −0.08 and 0.21 percentage points at every horizon and never significant. The 95 percent confidence intervals exclude declines larger than 0.25 percentage points three years after a change in the index and larger than 0.4 percentage points after five. Two one-sided equivalence tests reject cumulative effects larger than 0.5 percentage points in absolute value at horizons of one to four years, and at five years in the specification without controls. The minimum detectable effect is between 0.18 and 0.49 percentage points. For comparison, the baseline calibrations of Kaplan *et al.* (2020), Garriga & Hedlund (2020) and Castellanos *et al.* (2024) imply aggregate responses to changes in borrowing constraints of 1.8 to 3 percentage points. A one-unit change in the index is the activation of one instrument; the median tightening in the sample is 1.3 units, so the bounds for a typical tightening are about 30 percent wider.

The paper makes three contributions. The first is the aggregate result itself, together with an explicit account of what it can and cannot detect. Macroprudential policies act on new mortgages, and the homeownership rate is a stock dominated by incumbent owners, so a small aggregate response is compatible with a large effect on the marginal buyer. Section 10 makes this precise with a stock-flow benchmark. With roughly 1.3 percent of households entering ownership each year, a policy that permanently excludes 20 percent of first-time buyers lowers the aggregate rate by 1.3 percentage points over five years, which the estimates reject. The equivalence bound of 0.5 percentage points over five years corresponds to a permanent exclusion of about 8 percent of first-time buyers. The aggregate rate is therefore informative about persistent exclusion of this size or larger, and uninformative about temporary delays in entry.

The second contribution is evidence on the margins where the marginal buyer should be visible. Using EU-SILC tenure data by household composition and by age, I estimate the same local projections for the homeownership rate of adult-only households under 65, of individuals aged 18 to 54 and 25 to 54, and for the share of young households who own with a mortgage. The last series is a stock, but among the available cross-country series it is the one most exposed to first-time-buyer entry. None of these margins falls after a tightening. The young mortgagor share moves by less than 0.3 percentage points at every horizon, with no negative coefficient

that is distinguishable from zero. In models where landlords convert housing between segments, a credit tightening raises the share of households renting at market rates. In the data this share falls slightly, by 0.1 to 0.4 percentage points, and the estimates are imprecise.

The third contribution is to ask whether the borrower-based reforms studied in the micro literature are visible in national aggregates. I estimate the staggered difference-in-differences estimator of Callaway & Sant'Anna (2020) around the introduction of loan-to-value and loan-to-income limits in the Netherlands (2011), the United Kingdom (2014), Ireland (2015), Sweden (2016) and Finland (2016).[2] These reforms had large effects on the borrowers they targeted, but they leave no common trace in aggregate household credit, house prices or the aggregate homeownership rate. The share of young households owning with a mortgage is 1.3 percentage points lower after the reforms than in control countries, but it had been falling relative to controls by a similar amount before them, the estimate is not significant under permutation inference, and the reform-specific estimates differ in sign. The gap between the borrower-level and the aggregate evidence is informative in itself. The micro studies identify relative effects within a country, which difference out any common response, and they document that lenders reallocated credit toward unconstrained borrowers and that excluded buyers entered later. Effects estimated on treated borrowers therefore need not aggregate, and the same mechanisms would keep the aggregate homeownership rate from responding to the average tightening. Two-way fixed effects event studies, on which the earlier version of this paper and much of the literature rely, give estimates two to three times larger. With five treated countries these event studies are suggestive rather than decisive.

Two caveats shape how the results should be read. First, the macroprudential index has a weak first stage in this sample. Real house prices fall by about one percent three years after a one-unit tightening in the specification without controls, but the estimate is imprecise and not robust to controls, and aggregate credit does not respond on average. The five large borrower-based reforms in Section 9 do not supply a stronger first stage, since aggregate credit and house prices show no common movement around them either. The average conceals two groups of

[2] Van Bekkum *et al.* (2024) for the Netherlands, Peydró *et al.* (2024) for the United Kingdom, Acharya *et al.* (2022) for Ireland, Bäckman *et al.* (2024) for Sweden and Eerola *et al.* (2023) for Finland.

tightenings: after 51 of 127, real household debt fell over the following two years, while after the rest it kept rising, mostly in converging economies with deepening mortgage markets. The homeownership rate does not fall after either group, but this split conditions on an outcome observed after the policy and is descriptive. A split on a predetermined variable, the depth of the mortgage market before the policy change, shows that the tightenings that reduced credit were concentrated in countries with shallow mortgage markets; in those countries the homeownership rate falls by 0.2 to 0.4 percentage points in the specification without controls and does not fall in the specification with controls. Second, the paper does not identify the mechanism behind the results. Section 2 uses the segmented-market framework of Greenwald & Guren (2025) and the integrated-market framework of Kaplan *et al.* (2020) to organize the predictions, and Table 1 lists the margins on which they differ. The evidence is consistent with the segmented framework on every margin and hard to reconcile with the aggregate magnitudes of the integrated framework's central calibrations. A null response is also consistent with an index that has little effect on credit, with delayed rather than prevented entry, with exemptions that protect marginal buyers, and with effects that differ in sign across countries.

The paper contributes to the literature on credit and housing markets. Many papers estimate the causal effect of credit and interest rates on house prices (Glaeser *et al.* , 2012; Adelino *et al.* , 2025; Favara & Imbs, 2015; Loutskina & Strahan, 2015; Di Maggio & Kermani, 2017; Mian & Sufi, 2019), but few examine homeownership. Greenwald & Guren (2025) show for US regions that credit shocks move house prices but not homeownership rates, and interpret the result through segmentation between rental and owner-occupied housing. The results here extend that finding to a cross-country setting and to a different source of variation in credit conditions. The paper also contributes to the empirical literature on macroprudential policy summarized in the footnote above, which has established effects on credit and house prices but not examined the homeownership rate, and to a growing body of micro evidence on borrower-based measures (Acharya *et al.* , 2022; Peydró *et al.* , 2024; Van Bekkum *et al.* , 2024; Eerola *et al.* , 2023; Bolliger *et al.* , 2025; Aastveit *et al.* , 2023; Tzur-Ilan, 2023). That literature documents effects on transactions, leverage and the timing of purchases. The aggregate evidence here shows that

these effects have not translated into changes in who owns.

The results speak to policy in a limited way. They indicate that recorded tightenings of macroprudential regulation have not been followed by declines in aggregate homeownership over horizons of up to five years, so the case for relaxing borrower-based measures cannot rest on the homeownership rate. They do not show what loosening would do: only 18 of the 161 non-zero changes in the index are loosenings, and their coefficients are too imprecise to be informative. The results are also silent on the welfare of the marginal buyer, on delays in entry, on household formation, and on other costs of macroprudential regulation, such as reductions in household liquidity (Aastveit *et al.* , 2023).

The rest of the paper proceeds as follows. Section 2 sets out the conceptual framework and the predictions tested. Section 3 describes the data. Section 5 presents the empirical strategy. Section 6 reports the first stage and the aggregate results, Section 7 the results for young households and the flow into mortgaged ownership, Section 8 the rental market, and Section 9 the event studies around large reforms. Section 10 translates the bounds into the flow of first-time buyers, Section 11 reports robustness checks, and Section 12 concludes.

2 Conceptual framework

Two modeling traditions make different predictions about how a credit tightening affects the homeownership rate. In the first, the markets for owner-occupied and rental housing are segmented, so a credit shock moves house prices but leaves the homeownership rate largely unchanged (Greenwald & Guren, 2025). In the second, deep-pocketed landlords stand ready to convert housing between the rental and owner-occupied segments, so a credit shock moves both prices and the homeownership rate (Kaplan *et al.* , 2020). This section summarizes the two traditions and derives the predictions that the empirical sections examine. The aim is not to adjudicate between the models. A cross-country null is consistent with several mechanisms. The aim is instead to be explicit about the magnitudes each tradition predicts, so that the equivalence bounds estimated below can be read against them.

2.1 Segmented markets

The framework closely follows Greenwald & Guren (2025), who present a simple supply and demand framework for how changes in credit affect the homeownership rate.

Imagine that there is a market for owner-occupied housing but that households could also choose to rent. Following the literature using user-cost models (Poterba, 1984; Himmelberg *et al.* , 2005), the price of owning will be determined by an arbitrage relationship between the cost of renting and the cost of owning, commonly referred to as the user cost. Prices will adjust to equalize the user cost of owning and renting, $P = r/u$, where P is the price of housing, r is the rental price and u is the user cost. We can rewrite this as $P/r = 1/u$, giving us that the price-to-rent ratio equals the user cost. As the price-to-rent ratio increases, more households prefer to rent than to own, which creates a downward-sloping demand for ownership. Assuming that households require credit to own housing—a reasonable assumption given that most households use credit to purchase housing or are competing with other buyers who require credit—a reduction in credit supply will shift the demand for housing inward.

Note that this model says nothing about the homeownership rate, only about the equilibrium price-to-rent ratio. The ownership rate could in principle range from 0 to 100 percent, depending on the relative cost of owning and renting. In principle, if house prices adjust more rapidly than the supply of owner-occupied housing, the market can reach a new equilibrium where the price-rent ratio adjusts to maintain the user cost equality. In addition, rental markets could also adjust to maintain the user cost equality (Castellanos *et al.* , 2024). In these scenarios, the homeownership rate remains unchanged because the relative cost of owning versus renting is preserved through price adjustments rather than quantity changes. This would imply that the supply curve is perfectly inelastic. This mechanism explains why macroprudential policies can affect house prices without necessarily altering the extensive margin of homeownership participation.

The supply curve is not the general supply curve for housing, but the supply curve for owner-occupied housing relative to renting. Intuitively, the slope of the supply curve comes either from

new construction of owner-occupied housing or from landlords or owners who decide to convert rental units to owner-occupied housing. The slope of the supply curve therefore represents (a) the willingness of construction companies to expand the share of owner-occupied housing in their total construction when prices increase and (b) the willingness of landlords to convert existing supply from rental to owner-occupied housing.

What will be the impact of a change in demand on prices and the homeownership rate? As in standard supply and demand frameworks, the answer depends on the slope of the supply curve. If the supply curve is perfectly inelastic, the price will increase by the same amount as the demand shock. If the supply curve is perfectly elastic, the price will not change. If the supply curve is somewhere in between, the price will change by less than the demand shock. The slope of the supply function therefore determines the response of both the homeownership rate and house prices to a change in credit. Previous studies have found that macroprudential policies affect both credit and house prices. This suggests that we are not in the world where the supply of owner-occupied housing is perfectly elastic: if supply were perfectly elastic, macroprudential policies could affect credit but not house prices. In the segmented case, prices adjust and quantities do not. The rental market is affected only through the price-to-rent ratio, and the composition of the rental stock, in particular the share of households renting at market rates, does not need to change.

Greenwald & Guren (2025) show that the slope of the supply curve maps to different assumptions in macroeconomic models. In Greenwald (2017) and Favilukis *et al.* (2017), households cannot rent housing, which creates a perfectly inelastic supply curve where any demand shock is fully reflected in prices. The empirical evidence in Greenwald & Guren (2025) for US regional markets points in the same direction: credit shocks move house prices, but not the homeownership rate.

2.2 Integrated markets with elastic landlords

The second tradition assumes that the rental and owner-occupied segments are integrated. In Kaplan *et al.* (2020), identical, risk-neutral and deep-pocketed landlords buy and sell housing,

and the price of housing is tied to the present value of future rents. When credit tightens, households who can no longer buy rent instead, landlords buy the units they vacate, and rents and prices fall together by less than they would in a segmented market. In terms of the framework above, the supply of owner-occupied housing relative to renting is elastic, so a credit tightening lowers house prices by less and lowers the homeownership rate by more than in the segmented case, and it raises the share of households renting from landlords at market rates.

Models in this tradition generate quantitatively large responses. A relaxation of payment-to-income and loan-to-value constraints raises the homeownership rate by roughly 3 percentage points in Kaplan *et al.* (2020). A tighter loan-to-value limit lowers the homeownership rate by 2.8 percentage points in Garriga & Hedlund (2020). Castellanos *et al.* (2024) introduce endogenous landlords, which moves the model toward the segmented case, but still find that the 2016 loan-to-value and loan-to-income limits in Ireland lower the long-run homeownership rate by 1.8 percentage points. Balke *et al.* (2023) find large effects of down-payment requirements on the transition into homeownership. Because macroprudential policies act on new mortgages rather than on the existing stock, these models also predict that the response is concentrated among households at the margin of entry: young households, first-time buyers, and households who need a mortgage to buy.

2.3 Predictions across margins

Table 1 summarizes the predictions of the two traditions for the margins that the data allow me to examine. Both traditions predict that a tightening lowers credit and house prices, although the price response is smaller with elastic landlords, so a first-stage response of these variables is a precondition for the remaining tests rather than a discriminating prediction. Section 6.1 shows that this precondition is not met for the average change in the index, and Section 9 that it cannot be verified for the five large reforms. The null results below are therefore read as bounds on the aggregate effect of recorded policy actions rather than as a test between the two traditions. The traditions differ on the aggregate homeownership rate, on the ownership rate of young

Table 1: Predicted responses to a macroprudential tightening

Margin	Segmented markets	Integrated markets	Evidence in
House prices	Fall	Fall, by less	Section 6.1,
Credit	Fall	Fall	Section 6.1,
Aggregate homeownership rate	No response	Fall by 1.8–3 pp	Section 6
Young-household ownership rate	No response	Larger fall than aggregate	Section 7
Young households owning with a mortgage	No response	Fall	Section 7.2
Share renting at market rates	No response	Rise	Section 8

Notes: Segmented markets refers to the class of models in Greenwald & Guren (2025), Greenwald (2017) and Favilukis *et al.* (201 owner-occupied housing relative to rental housing is inelastic. Integrated markets refers to the class of models with elastic landlord co (2020), Garriga & Hedlund (2020), Castellanos *et al.* (2024) and Balke *et al.* (2023). The quantitative range for the aggregate homeow the baseline calibrations in those papers.

households, on the share of young households who own with a mortgage, which is the available series most exposed to first-time-buyer entry, and on the composition of the rental market. The integrated tradition predicts responses on all four margins, and the calibrations above put the aggregate response at 1.8 to 3 percentage points. The segmented tradition predicts no response on any of them.

The empirical strategy follows from this table. Section 6.1 asks whether the policy index moves credit and house prices in this sample, and Section 9 asks the same of the five large reforms. Section 6 estimates the aggregate response and reports equivalence bounds. Sections 7 to 8 examine the margins on which the integrated tradition predicts the largest responses. Section 10 translates the aggregate bounds into the flow of first-time buyers, which is the object the structural models discipline.

3 Data and variables

The empirical analysis combines homeownership and tenure data from the European Union Statistics on Income and Living Conditions (EU-SILC), macroprudential policy actions from the Macroprudential Policies Evaluation Database (MaPPED) provided by the European Central Bank (Budnik & Kleibl, 2018), macroeconomic, house price, debt and interest rate series from Eurostat, and credit and policy rate data from the Bank for International Settlements (BIS). All data are collected or collapsed to the annual level to match the frequency of the homeownership

data. Below I describe the homeownership and macroprudential data and the criteria used for selecting time periods and countries. The macro data are described in Appendix C.

3.1 Homeownership and tenure data

I collect homeownership data from EU-SILC. Aggregate homeownership data are available for 35 countries, with the earliest series starting in 2003. In addition to the aggregate homeownership rate, I collect the share of households that own their home with or without a mortgage, the share renting at market rates and at reduced rates or rent-free, and all categories for households above or below 60 percent of median equivalized income. In the empirical analysis, I use these shares in levels, in percentage points of all households.

To examine the marginal buyer, I use two further EU-SILC breakdowns. The first is tenure by household composition, which separates adult-only households in which all members are under 65 from adult-only households with at least one member aged 65 or over. For each group I compute the share owning, the share owning with a mortgage, the share owning outright, and the share renting at market rates and at reduced rates. Households with dependent children are not included in either group, so the "young" group consists of singles and couples without children under 65. The second breakdown is tenure by age of the individual, which provides ownership rates for individuals aged 18 to 54, 25 to 54, 55 to 64 and 65 and over. Both breakdowns are available for the same 28 countries and years as the aggregate series. The share of young adult-only households owning with a mortgage is a stock, not a flow. It rises when a young household buys with a mortgage, and it falls when mortgagors repay, age past 65, have children and leave the group, or sell. Among the available cross-country series it is the one most exposed to first-time-buyer entry, and I use it as a proxy for that margin with these limitations in mind. It also excludes families with dependent children, who account for a substantial share of first-time buyers.

3.2 Macroprudential policies

I use the Macroprudential Policies Evaluation Database to construct a country-specific macroprudential policy index. The database contains 1,417 policy actions undertaken in 28 countries after 1995.[3] The database covers 11 policy instruments including capital buffers, lending standards restrictions, leverage ratios, and liquidity requirements. The last update was the fourth quarter of 2017, which marked the end of the sample period.

The MaPPED data have several advantages compared to other datasets (Fernández-Gallardo *et al.* , 2025). The data are based on surveys of national regulators, allowing comparison of policy actions across countries. The data contain information about each policy action, including announcement and implementation dates, and whether the policy involved tightening, loosening or ambiguous effects. Importantly for identification, the survey asks if the policy had a countercyclical motivation or design.

Following Fernández-Gallardo *et al.* (2025) and Fernandez-Gallardo (2023), I construct an overall index of macroprudential policies using all non-countercyclical policies. I use the announcement date to assign a value to each policy, using a positive value for tightening, a negative value for loosening and a zero for ambiguous actions.[4] Following (Meuleman & Vander Vennet, 2020), I account for differences in scope between new and existing policies by assigning different weights. The weighting scheme gives higher weight to activations and deactivations, compared to changes in scope or level. The weighting scheme is provided in Table B1.

For each country, I then construct a series as the weighted average of macroprudential policy actions active for a given year. The variable of interest is then the year-over-year change in the index, denoted $\Delta MaPP_{i,t}$.

[3] The database contains several policies enacted before 1995, if they are still in place in 1995. In total, the database contains 1,925 policies.

[4] The implementation date is also noted in the dataset. The median difference between announcement and implementation is 1 month, and the average difference is 5 months.

3.3 Sample selection and summary statistics

The sample selection is guided by data availability for homeownership rates and macroprudential policies. The final sample includes 28 countries. Homeownership data begins in 2003, but with many missing observations. The start and end dates for all countries with available homeownership data are listed in Table B2, along with homeownership rates in the first year, last year and on average for all years.

4 Homeownership rates and macroprudential policies, 2010–2017

Homeownership rates move slowly. Figure 1 plots the homeownership rate in 2017 against the rate in 2010 for each country, and Appendix Figure A1 plots the full time series. The rate declined in most countries between 2010 and 2017, by 1.25 percentage points on average, and the decline was larger in the countries that used fewer macroprudential policies. A regression of the change in homeownership on the change in the macroprudential index between 2010 and 2017 gives a positive and insignificant coefficient of 0.2. Cross-country differences in levels are large and persistent: rates above 80 percent in Romania, Lithuania and Slovakia reflect the sale of state housing to tenants after 1990 (Broulíková *et al.* , 2020), and rates near 50 percent in Germany and Austria reflect rental market institutions and preferences (Kaas *et al.* , 2021; Huber & Schmidt, 2022). These differences are absorbed by country fixed effects in the analysis below; the persistence of the series is why the empirical design focuses on changes over horizons of up to five years.

5 Empirical strategy

This section describes the empirical strategy designed to identify how changes in macroprudential policies affect the homeownership rate in aggregate and for different groups. The analysis follows the literature on macroprudential policies (Fernández-Gallardo *et al.* , 2025; Fernandez-Gallardo, 2023; Kuttner & Shim, 2016; Cerutti *et al.* , 2017). I use a narrative approach to

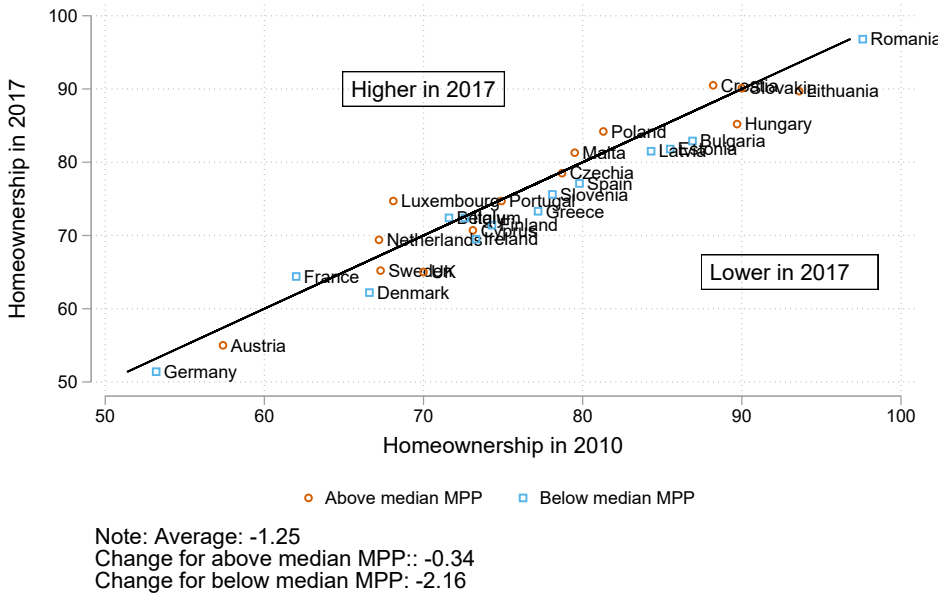


Figure 1: Change in homeownership rates 2010-2017 and macroprudential policies

Notes: The figure shows the homeownership rate in 2010 on the x-axis and the homeownership rate in 2017 on the y-axis. The color of the points indicates if the country had an above-median number of macroprudential policies since 2010. The solid line marks the 45-degree line, showing whether homeownership rates increased or decreased between 2010 and 2017.

identify macroprudential policy shocks, and a local projection (LP) framework to estimate the effect of macroprudential policies on the homeownership rate.

5.1 Identifying macroprudential policy shocks

To identify the effect of macroprudential policies on the homeownership rate, it is necessary to measure only exogenous components of macroprudential policy. In this setting, that involves identifying policies that do not systematically respond to changes in the homeownership rate or economic fluctuations that may drive changes in the homeownership rate. Overall, this is a challenge that arises because macroprudential policies are not introduced randomly, but instead often respond to the state of the economy. For example, macroprudential policies may be introduced in response to overheating of the economy. If policy tightening occurs at housing market peaks, the homeownership rate could decline for reasons unrelated to the policy.

To address this challenge, I use the narrative approach of Fernández-Gallardo *et al.* (2025)

to identify macroprudential policy actions that do not have a counter-cyclical motive. The MaPPED database contains a variable about the motivation for introducing a specific policy action. I use this variable to identify policies that do not have a counter-cyclical motive. A measure has a countercyclical design when the instrument automatically tightens when systemic risk intensifies and loosens when systemic risk decreases (Budnik & Kleibl, 2018). I remove all policy actions with this characteristic. The main variable of interest represents a shock to the aggregate macroprudential policy stance for each country.

A common approach in the literature is to include variables that could motivate the introduction of macroprudential policies, such as credit growth or house prices. The narrative approach addresses this concern by excluding counter-cyclical policies from the outset. However, macroprudential policies may still be correlated with the state of the economy. To address this concern, I include control variables for the growth of private sector debt relative to GDP, the change in the real house price index, and the change in the short-term interest rate, alongside GDP growth per capita and inflation. I use two sets of credit and interest rate series. The first, used in the previous version of this paper, comes from the BIS and is unavailable for ten of the 28 countries, which cuts the with-controls sample by more than 40 percent. The second comes from Eurostat and covers all 28 countries. Appendix Table B9 reports the estimation sample under each set, and Appendix C describes the series. The two sets give very similar results, and I report the Eurostat set as the baseline with controls.

5.2 Threats to identification

There are several other threats to identification, beyond the potentially endogenous implementation of macroprudential policies. I now discuss these in more detail.

Omitted variable bias. If macroprudential policy moves together with the macro and housing cycle, there may be an omitted variable bias. For instance, loose monetary policy could increase housing demand or change expectations over future house prices, which may influence both the homeownership rate and the introduction of macroprudential policies. To address this concern, I include a number of control variables.

Measurement error. Measurement error in policy indicators may mean that I assign a high weight to a policy that has a limited impact or scope. To address this concern, note that the weighting scheme of the MaPPED database allows me to give a lower weight to policies that have a limited impact or scope. In robustness checks, I also use another macroprudential policy database, and run results using no weighting. In unreported results, I find no differences in results.

Anticipation effects. The MaPPED database contains information about both the announcement and implementation of macroprudential policies. If macroprudential policies are announced and implemented at different times, it may be the case that the implementation of a policy does not affect the homeownership rate, but the announcement does. To address this concern, I use the announcement date as the date of the policy. Overall, the difference between the announcement and implementation date is small, with an average difference of five months and a median difference of one month. Since I use annual data, the difference between the announcement and implementation date is unlikely to be a major concern.

Data frequency. Statistical power is potentially a concern with annual data. However, unlike financial variables such as credit growth or house prices, which can respond quickly to policy changes, homeownership rates reflect long-term household decisions about tenure choice that are typically made infrequently and with considerable deliberation. Even if macroprudential policies do affect the attractiveness of homeownership, these effects may take several years to materialize as households gradually adjust their housing tenure decisions. For this reason, using annual data is appropriate in this setting.

5.3 Empirical specification

I investigate the relationship between homeownership and macroprudential policies using local projections (LP) following Jordà (2005), Richter *et al.* (2019) and Coulier *et al.* (2022). I use this methodology as it enables me to estimate the effect of macroprudential policies over time, which is particularly relevant for a slow-moving variable such as homeownership rates. I

estimate the following baseline model:

$$\text{Owner}^{j}_{i,t+h-1} - \text{Owner}^{j}_{i,t-1} = \alpha^{h}_{i} + \gamma^{h}_{t} + \beta_{h}\Delta MaPP_{i,t} + \phi^{h}\Delta X_{i,t-1} + \epsilon_{i,t+h-1}, \tag{1}$$

for $h = 1, \ldots, 5$. The dependent variable is the cumulative change in the homeownership rate for country i and group j from the year before the policy change to $h-1$ years after it, so that β_1 is the response in the year of the change and β_5 the level of the rate four years later relative to the year before. All estimates in the paper are cumulative responses of this kind; an earlier version estimated year-on-year changes at each horizon, which cannot be compared to the level predictions of the structural models or to the stock-flow benchmark in Section 10. The variable of interest, $\Delta MaPP_{i,t}$, is the change in the cumulative index of macroprudential actions. I include a vector of controls in $X_{i,t-1}$. Control variables are all lagged one year, and include GDP growth per capita, inflation, real house price growth, growth in private sector debt relative to GDP, the change in the short-term interest rate, and the level of the macroprudential index. α^{h}_{i} and γ^{h}_{t} are country and time dummies that account for cross-country differences in the average homeownership rate and for common time trends. Standard errors are clustered by country; Section 11 reports wild cluster bootstrap, Driscoll-Kraay and randomization inference.

A one-unit change in the index corresponds to the activation of one new instrument, or to four changes in the level of existing instruments. In the estimation sample the index change has a standard deviation of about 1.5 and the median non-zero tightening is 1.35, so the tables also report the effect of a one-standard-deviation change and of the median tightening. The index does not measure the intensity of a restriction: an activation of a capital buffer and an activation of a loan-to-value limit enter with the same weight. Two features of the index matter for reading the results. First, the index change is positive in 143 country-years and negative in only 18, so the estimates are driven by tightenings; Section 11 estimates tightenings and loosenings separately. Second, because actions coded as counter-cyclical in MaPPED are excluded, several of the reforms studied in the micro literature do not register: the Irish 2015 limits and the Finnish 2016 cap enter the index as zero, the Swedish 2016 amortization requirement as 0.25, the UK 2014 loan-to-income limit as 2.1 and the Dutch 2011 loan-to-value cap as 3. Appendix Table

B8 lists every non-zero change in the index. The narrative index therefore measures recorded changes in the regulatory stance, and the results should be read as the average response to such changes rather than as the response to a credit shock of known size. Section 9 examines the large reforms directly.

I examine several measures of homeownership. All measures are expressed in percentage points of the relevant population. First, I use the aggregate homeownership rate. Second, I examine households with and without a mortgage, a distinction that is particularly relevant for macroprudential policies that directly target mortgage lending, and households above and below 60 percent of median equivalized income. Third, I examine young households and the share of young households owning with a mortgage, and fourth the composition of the rental market. For the first-stage regressions in Section 6.1, the dependent variable is the cumulative change in the log of house prices or credit between $t-1$ and $t+h$, so that the coefficients are percent changes.

Alongside conventional tests, I report equivalence tests and minimum detectable effects. A null result is only informative if the confidence interval is narrow relative to the effects of interest. The two one-sided tests procedure (Lakens, 2017) rejects the hypothesis that the effect lies outside $[-\delta, \delta]$ at the 5 percent level when the 90 percent confidence interval lies inside that interval. I report the test at $\delta = 0.5$ percentage points in the main tables and at $\delta \in \{0.1, 0.25, 0.5, 1\}$ in Appendix Table B11, together with the minimum effect that would be detected with 80 percent power. Section 10 maps each margin into the share of first-time buyers that a policy would need to exclude to produce it.

6 Aggregate results

6.1 First stage: house prices and credit

Both frameworks in Section 2 predict that a macroprudential tightening lowers credit and house prices, and the literature has documented both responses (Cerutti *et al.* , 2017; Kuttner & Shim, 2016; Alam *et al.* , 2019; Fernández-Gallardo *et al.* , 2025). Before turning to homeownership,



Figure 2: First stage: response of real house prices and private debt to the macroprudential index

Notes: Cumulative response, in percent, of the real house price index (left) and of private sector debt relative to GDP (right) to a one-unit change in the macroprudential index, $y_{i,t+h} - y_{i,t-1}$ regressed on $\Delta MaPP_{i,t}$ with country and year fixed effects. The blue line excludes and the red line includes control variables (lagged GDP growth per capita, inflation, the change in the 3-month money market rate, the level of the index, and the lagged growth of the other variable). 28 countries. 95 percent confidence intervals in shaded areas, clustered by country.

I therefore ask whether the narrative index moves these variables in this sample. Figure 2 plots the cumulative response of real house prices and of private sector debt relative to GDP to a one-unit change in the index, with and without controls. The coefficients are in Appendix Table B12, which also reports household debt and the BIS credit series.

The first stage is weak. Real house prices fall after a tightening in the specification without controls, by about one percent after three years and 1.4 percent after five, but only the three-year estimate is significant at the 10 percent level, and the response disappears once controls are included. Private sector debt relative to GDP does not respond at any horizon; the point estimates are positive and small. Household debt and the BIS credit series fall by about 1.5

percent after five years in the specifications with controls, significant at the 10 percent level, and not before. Restricting the shock to lending standard restrictions, the borrower-based instruments that act most directly on mortgage credit, changes the estimates only a little (Appendix Table B13 and Figure A7). Real house prices fall by between 0.5 and 1.8 percent over the five horizons without controls, but no estimate is significant, and household debt relative to GDP falls by 2.2 percent after five years, significant at the 5 percent level. Private sector debt relative to GDP, which includes corporate debt, rises after a tightening of lending standards, which is consistent with borrower-based measures being adopted during credit expansions that continue for some years.

The positive point estimates for private debt reflect heterogeneity across tightenings rather than reverse causality. Past household credit growth does not predict tightenings of lending standards once country and year effects are included. Instead, the tightenings split into two groups. After 51 of the 127 tightenings of the overall index that can be classified, real household debt fell over the following two years; after the other 76 it kept rising. The second group is dominated by converging economies (Slovakia, Slovenia, Poland and Czechia) and by Sweden, where borrower-based measures were introduced during a structural deepening of mortgage markets that continued for years. The first group includes Hungary, the Baltic states, Croatia and the Netherlands, where tightenings coincided with the end of credit booms. Averaging across the two groups produces a first stage close to zero.

Two readings of the weak average first stage are possible. The narrative index measures the stance of regulation, and the average non-counter-cyclical action in MaPPED may be too small to move aggregate credit and prices in annual data across 28 countries. Alternatively, the effects documented in the literature are concentrated in the tightenings that reduced credit, and the index also counts many measures that did not. The rest of the paper addresses this in two ways. Section 6.3 re-estimates the homeownership response using only the tightenings after which credit fell. Section 9 examines the large borrower-based reforms directly, where the first stage is established in the micro literature.

6.2 The aggregate homeownership rate

Changes in the macroprudential index are not followed by detectable changes in the aggregate homeownership rate at any horizon. Figure 3 plots the cumulative responses from Equation (1) without controls, with the Eurostat control set that covers all 28 countries, and with the BIS control set that covers 18. Table B10 reports the coefficients. Without controls, the cumulative response ranges from −0.08 to 0.04 percentage points, with standard errors between 0.08 and 0.16 that widen with the horizon. With the full-coverage controls, it ranges from 0.05 to 0.21 percentage points, with standard errors between 0.07 and 0.17; no coefficient is significant. The lower bounds of the 95 percent confidence intervals are −0.25 percentage points at three years and −0.39 at five years in the specification without controls, and −0.13 at five years with controls. The BIS control set on 18 countries gives positive coefficients that reach 0.3 and 0.5 percentage points at four and five years and are significant. In no specification does the homeownership rate fall. Table B3, the specification with controls used in the earlier version of the paper, now uses the full-coverage controls and coincides with Panel B.

The estimates are precise enough to be informative. Table B10 reports two one-sided equivalence tests at $\delta = 0.5$ percentage points and the minimum detectable effect at each horizon. Without controls, equivalence holds at every horizon; with the full-coverage controls it holds at horizons one to four and the p-value at five years is 0.051. The minimum detectable effect, the true cumulative effect that would be found with 80 percent power, is between 0.18 and 0.30 percentage points at horizons one to four and 0.44 to 0.49 percentage points at five years. Appendix Table B11 varies the margin. Equivalence at $\delta = 0.25$ percentage points holds at horizons one to three without controls and one to two with controls, and fails at longer horizons; equivalence at $\delta = 0.1$ fails everywhere. The estimates therefore bound the cumulative effect of a one-unit change in the index at half a percentage point over four years, and at a quarter of a percentage point over two to three years. Because the median tightening is 1.3 units, the corresponding bounds for a typical tightening are 0.65 and 0.33 percentage points. The baseline models in Garriga & Hedlund (2020) and Kaplan *et al.* (2020) predict responses of the homeownership rate to changes in loan-to-value and payment-to-income limits of 2.8 and 3 percentage points.

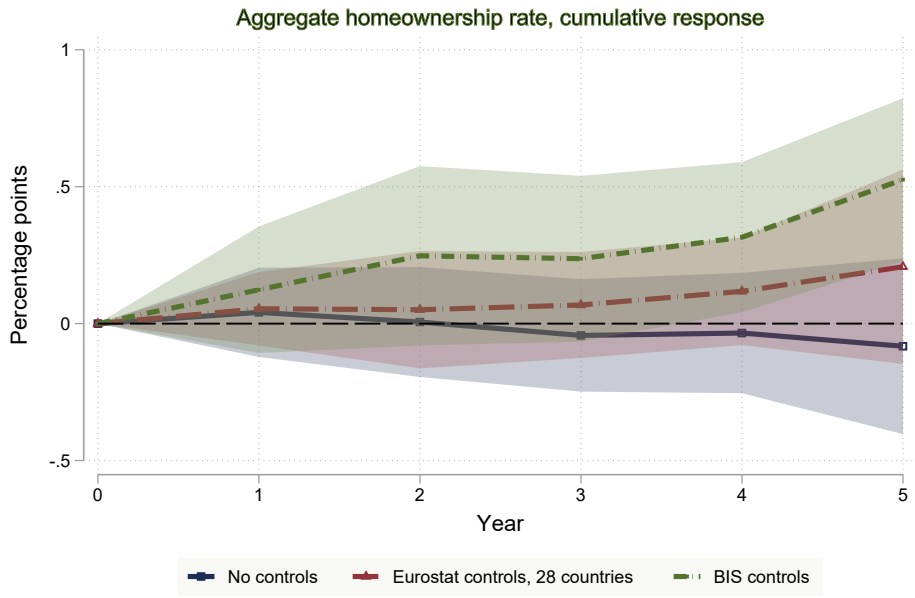


Figure 3: Cumulative response of the aggregate homeownership rate to the macroprudential index

Notes: Cumulative change in the aggregate homeownership rate from the year before a one-unit change in the macroprudential index, in percentage points, estimated using Equation (1). Year 1 is the year of the index change. The blue line includes country and year fixed effects only (28 countries). The red line adds lagged GDP growth per capita, inflation, real house price growth, growth in Eurostat private sector debt to GDP, the change in the 3-month money market rate and the level of the index (28 countries). The green line uses the BIS credit and policy rate series instead (18 countries). 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Table B10.

The bounds are several times smaller than these predictions, but the comparison is inexact because the model experiments are specific changes in borrowing limits and the index change is a count of regulatory actions.

The absence of a decline in the aggregate homeownership rate stands in contrast to the effects of macroprudential policies on credit and house prices documented in the literature. Where those policies have effects, they are on the volume of credit and the pace of price increases. The share of households who own does not move.

6.3 Tightenings that reduced credit

The weak average first stage leaves open whether the homeownership rate fails to respond to credit, or whether the index fails to move credit. To separate the two, I split each tightening of

the index into two components according to what happened to real household debt afterwards. A tightening is classified as effective if real household debt fell between the year before and two years after the change in the index, and as not effective otherwise; tightenings in the last two sample years cannot be classified and are dropped. The two components enter Equation (1) as separate regressors, so that each coefficient compares country-years with that type of tightening to country-years without one. The classification conditions on an outcome observed after the policy, so it describes what happened to homeownership when credit did fall rather than identifying a causal effect of effective policy. A second definition, which classifies a tightening as effective when household debt growth decelerated relative to the three preceding years, gives similar results.

Figure 4 and Appendix Tables B20 and B21 report the results. The tightenings after which credit fell have a first stage of the size the literature documents. Real household debt is 1.1 percent lower after two years and 2.1 percent lower after four; the decline in the first three years is partly by construction, but it persists to the fifth year, when it is no longer constrained by the classification. Real house prices, which play no role in the classification, fall by 1.6 percent after two years and 2.2 percent after three, significant at the 5 and 1 percent levels. The tightenings after which credit did not fall are followed by rising debt and flat house prices. The homeownership rate does not fall after either component. After the tightenings that reduced credit, the cumulative response is 0.13 to 0.16 percentage points without controls, not significant, and 0.17 to 0.29 percentage points with the full-coverage controls, significant at the 5 or 10 percent level at horizons one to four. The homeownership rate rose slightly in the years after these tightenings, which is the opposite of exclusion. The lower bounds of the 95 percent confidence intervals for this component are above -0.2 percentage points at every horizon. The share of young households owning with a mortgage and the share renting at market rates do not fall after these tightenings either. Appendix Table B22 and Figure A8 repeat the exercise for lending standard restrictions alone: real household debt falls by 1.2 to 3.5 percent after the 14 borrower-based tightenings that were followed by a credit decline, and the cumulative response of the homeownership rate is between -0.11 and 0.19 percentage points.

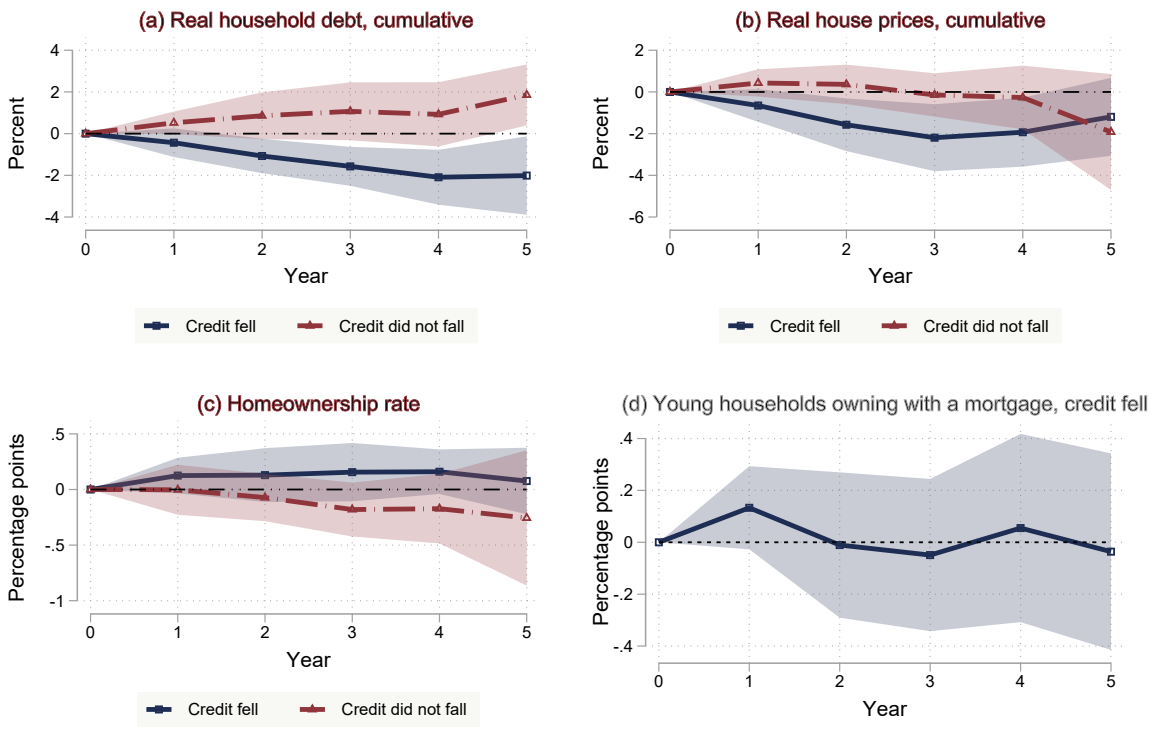


Figure 4: Tightenings followed by a fall in household credit: credit, prices, ownership

Notes: Responses to tightenings of the macroprudential index after which real household debt fell between $t-1$ and $t+2$ (blue) and to tightenings after which it did not (red), from local projections with country and year fixed effects in which both components enter jointly. Panels (a) and (b): cumulative percent change in real household debt and real house prices. Panel (c): change in the aggregate homeownership rate in percentage points. Panel (d): change in the share of adult-only households under 65 owning with a mortgage, for tightenings after which credit fell. 51 of 127 classifiable tightenings are followed by a fall in credit. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Appendix Tables B20 and B21.

Because this split conditions on an outcome, I also split the tightenings on a variable fixed before the policy change: household debt relative to GDP in the previous year, above or below the sample median of 52 percent. Deep mortgage markets are where borrower-based limits are most likely to constrain buyers, but the split also separates Western European countries from most of the converging economies. Appendix Table B23 reports the results. Tightenings in countries with shallow mortgage markets are followed by declines in real household debt of 1.5 to 2.9 percent, while tightenings in deep markets are followed by increases; this is the same pattern as the outcome-based split, and it says that the tightenings that reduced credit were mostly in the converging economies. The homeownership response differs across the two groups. In

deep mortgage markets the cumulative response is positive, between 0.07 and 0.24 percentage points, and not significant. In shallow markets it is −0.20, −0.33 and −0.37 percentage points at horizons two to four without controls, significant at the 5 and 10 percent levels, and between −0.11 and 0.41 with controls, not significant. A continuous interaction of the index change with lagged household debt to GDP is positive and not significant. The specification without controls therefore shows a decline in homeownership of 0.2 to 0.4 percentage points after tightenings in shallow mortgage markets, and the specification with controls does not. This is the only place in the cross-country evidence where a negative response of the homeownership rate appears, it is concentrated in the group of countries where credit fell, and it is not robust to controls.

Taken together, the tightenings after which credit fell by about two percent were not followed by declines in the aggregate homeownership rate, and were followed by small increases in the specification with controls. In shallow mortgage markets, where these tightenings are concentrated, the specification without controls shows a decline of a few tenths of a percentage point. Neither pattern approaches the magnitudes the integrated framework predicts. Section 9 reaches the same conclusion from the five large borrower-based reforms, with the difference that credit and prices do not visibly move there. The two exercises are the paper's closest approach to a credit shock of known effect, and both are suggestive rather than decisive, since the first conditions on an outcome and the second rests on five treated countries.

6.4 Mortgagors, outright owners and income groups

Next, I examine homeownership rates for different groups of households. A concern for policymakers is that macroprudential policies affect some groups more than others, in particular low-income households and households who need a mortgage. Figure 5 shows the cumulative responses for low-income and high-income households and for households with and without mortgages, with the full-coverage controls. Table B4 reports the coefficients. No group shows a decline that is distinguishable from zero. The share of households owning with a mortgage moves by less than 0.14 percentage points in absolute value at every horizon, and equivalence at 0.5 percentage points holds at every horizon. The low-income results are less precise. Low-income

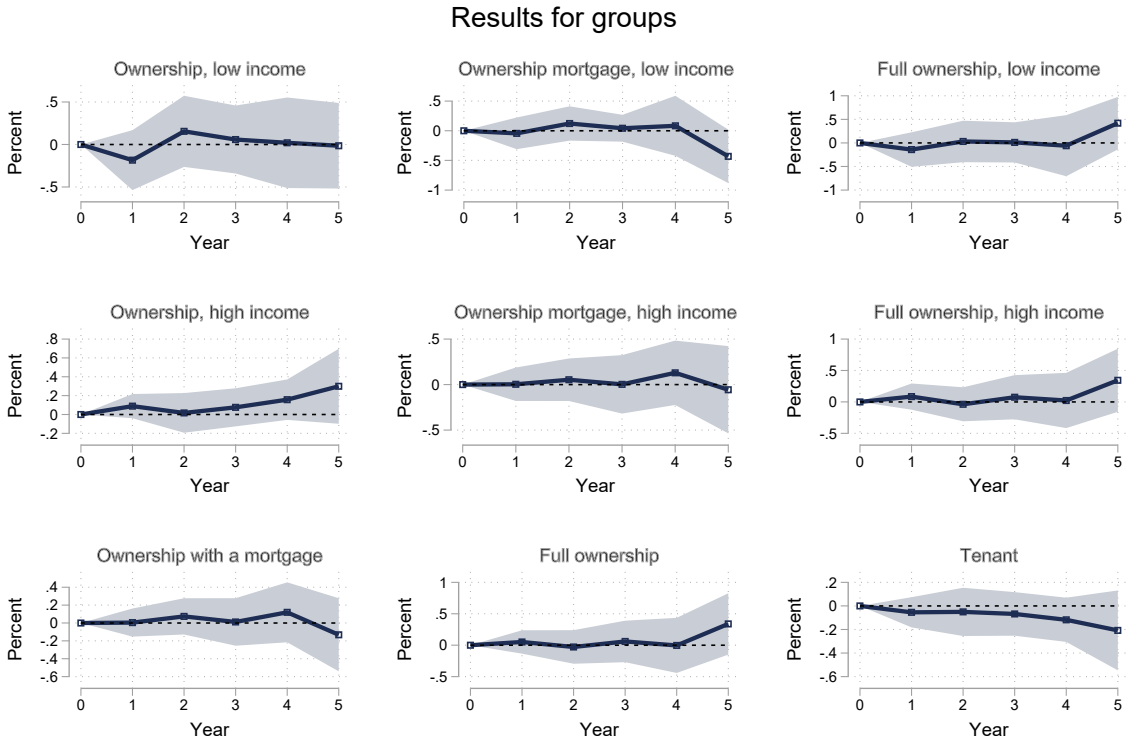


Figure 5: Cumulative response of homeownership rates by group to the macroprudential index

Notes: Cumulative change, in percentage points, in the homeownership rate of each group from the year before a one-unit change in the macroprudential index, estimated using Equation (1) with the Eurostat control set (lagged GDP growth per capita, inflation, real house price growth, growth in private sector debt to GDP, the change in the 3-month money market rate, and the level of the index). 28 countries. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Appendix Table B4.

households are defined by income below 60 percent of the median in the survey year, which is a small and changing group, and the standard errors are two to three times larger than for the aggregate rate. The cumulative response of low-income homeownership ranges from −0.18 to 0.16 percentage points, and the response of low-income mortgagors from −0.43 to 0.12, with the five-year coefficient of −0.43 significant at the 10 percent level. Equivalence at 0.5 percentage points fails at one horizon for low-income homeownership and at two horizons for low-income mortgagors. These estimates rule out large declines but not moderate ones, and they should not be read as precise nulls. Low income in the survey year is also not a measure of proximity to a borrowing constraint, since the group includes many retired outright owners.

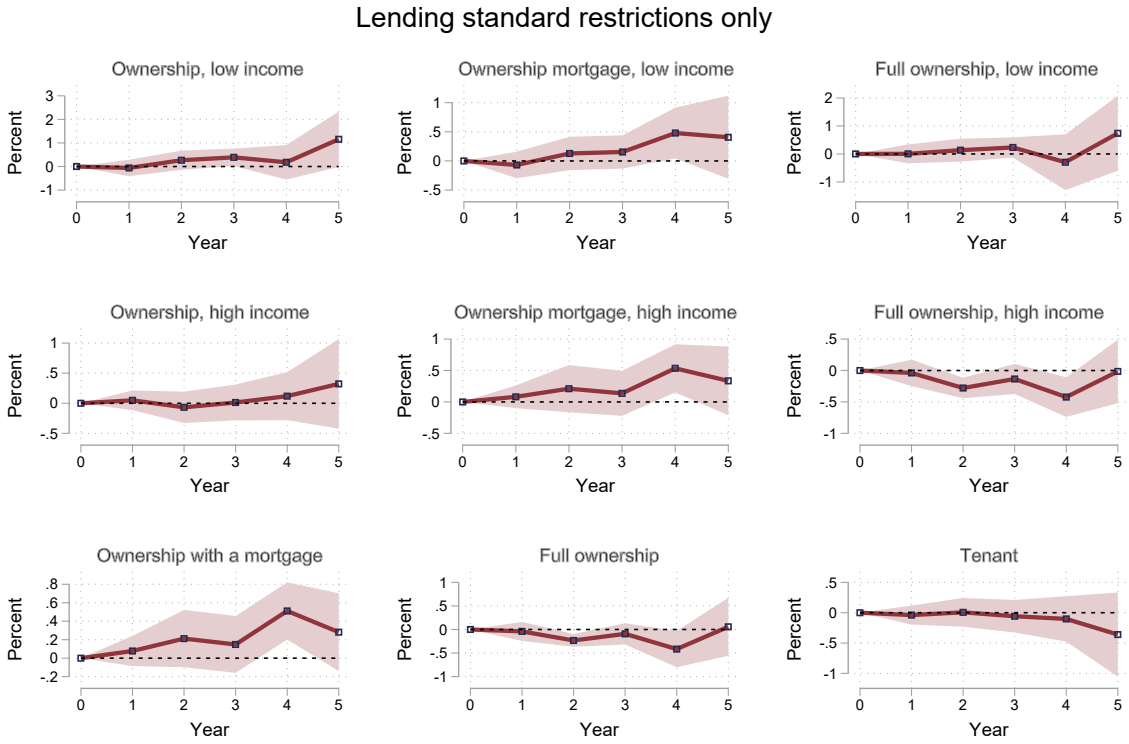


Figure 6: Cumulative response of homeownership rates by group to lending standard restrictions

Notes: Cumulative change, in percentage points, in the homeownership rate of each group from the year before a one-unit change in the index of lending standard restrictions, estimated using Equation (1) with the Eurostat control set. 28 countries. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Appendix Table B5.

6.5 Lending standard restrictions

The next set of results uses only lending standard restrictions. These restrictions, which include LTV and debt-service-to-income (DSTI) limits, are the macroprudential tools that act directly on household borrowing, and they are the tools the structural models study. I construct an index from these actions alone and estimate Equation (1) with it. The index has 28 non-zero tightenings in the estimation sample, so the estimates are less precise than for the overall index. Appendix Table B13 reports the cumulative response of the aggregate homeownership rate: between −0.38 and 0.17 percentage points without controls and between −0.01 and 0.36 with controls, with no coefficient significant and standard errors of 0.1 to 0.35. Figure 6 shows the responses by group. For low-income households the response is positive in the first year only. Lending standard restrictions are not followed by declines in homeownership at any horizon, but

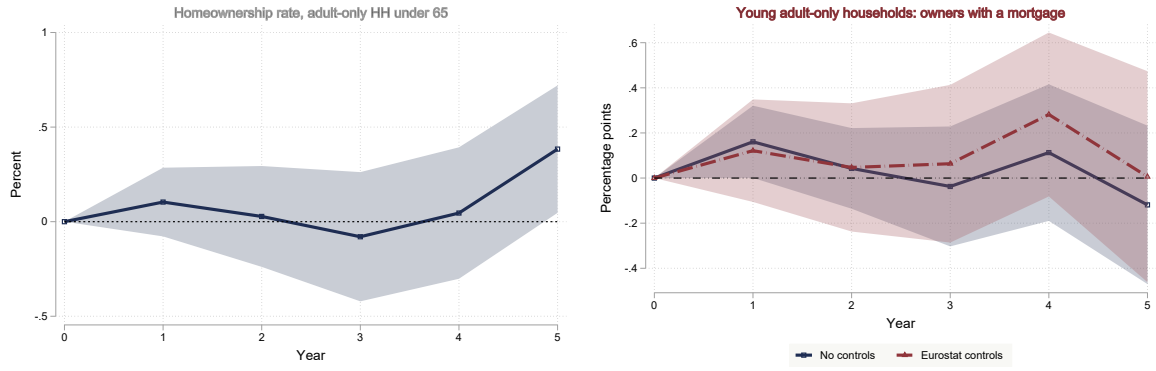


Figure 7: Young adult-only households: cumulative response of the homeownership rate and of the share owning with a mortgage

Notes: Cumulative change, in percentage points, in the homeownership rate of adult-only households under 65 (left) and in the share of those households who own with a mortgage (right) after a one-unit change in the macroprudential index, estimated using Equation (1). The blue line includes country and year fixed effects only; the red line adds the Eurostat control set. 28 countries. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Appendix Tables B16 and B18.

the confidence intervals for this index are roughly twice as wide as for the overall index.

7 Young households and the flow into mortgaged ownership

The aggregate homeownership rate is a stock dominated by incumbent owners, while macroprudential policies act on new mortgages. If the policies exclude marginal buyers, the effect should be visible first among young households and among households who need a mortgage to buy. This section examines both margins.

7.1 Homeownership of young households

The left panel of Figure 7 plots the cumulative response of the homeownership rate of adult-only households under 65, and Appendix Table B16 reports the coefficients. Without controls, the estimates range from -0.16 to 0.12 percentage points, none significant, with standard errors of 0.10 to 0.23. With controls, the estimates are positive except at three years, and the five-year coefficient of 0.38 is significant at the 5 percent level. Equivalence at $\delta = 0.5$ percentage points holds at every horizon without controls and at four of five horizons with controls. Appendix Table B17 and Figure A12 repeat the exercise for older adult-only households and for individuals

aged 18 to 54, 25 to 54, 55 to 64 and 65 and over. The cumulative responses for the two younger age bands are between 0.03 and 0.13 percentage points at every horizon; no age group shows a decline. The integrated-market framework predicts a larger decline for young households than in aggregate. Neither appears in the data. The age bands available in the Eurostat tables are broad; narrower groups such as 25 to 34 would require the EU-SILC microdata.

7.2 Young households owning with a mortgage

The right panel of Figure 7 turns to the share of young adult-only households who own with a mortgage. As discussed in Section 3, this is a stock that rises with first-time purchases and falls with repayment, ageing and changes in household composition; it is the available series most exposed to first-time-buyer entry, not a measure of the flow. Appendix Table B18 reports the coefficients. Without controls, the share is 0.16 percentage points higher in the year of a tightening, significant at the 5 percent level, and within 0.12 percentage points of its starting level thereafter. With the full-coverage controls, the cumulative responses are 0.12, 0.05, 0.06, 0.28 and 0.01 percentage points, none significant. There is no horizon at which the share is lower than before the tightening. Equivalence holds at $\delta = 0.5$ percentage points at nine of ten horizon-specification pairs and at $\delta = 1$ percentage point at all of them; the minimum detectable effect is between 0.22 and 0.64 percentage points. Since young mortgagors are about 21 percent of young households, a given change in first-time-buyer entry moves this share by more than it moves the aggregate rate, so the absence of a decline here is more informative than the aggregate result about the entry margin. The share of young households owning outright does not respond either (Appendix Figure A11).

8 The rental market

The two frameworks in Section 2 differ in how the rental market absorbs a credit tightening. With elastic landlords, households who can no longer buy rent instead, and landlords supply the units, so the share of households renting at market rates rises. With segmented markets, prices adjust and the composition of the rental stock is unchanged. Figure 8 brings the pieces together.

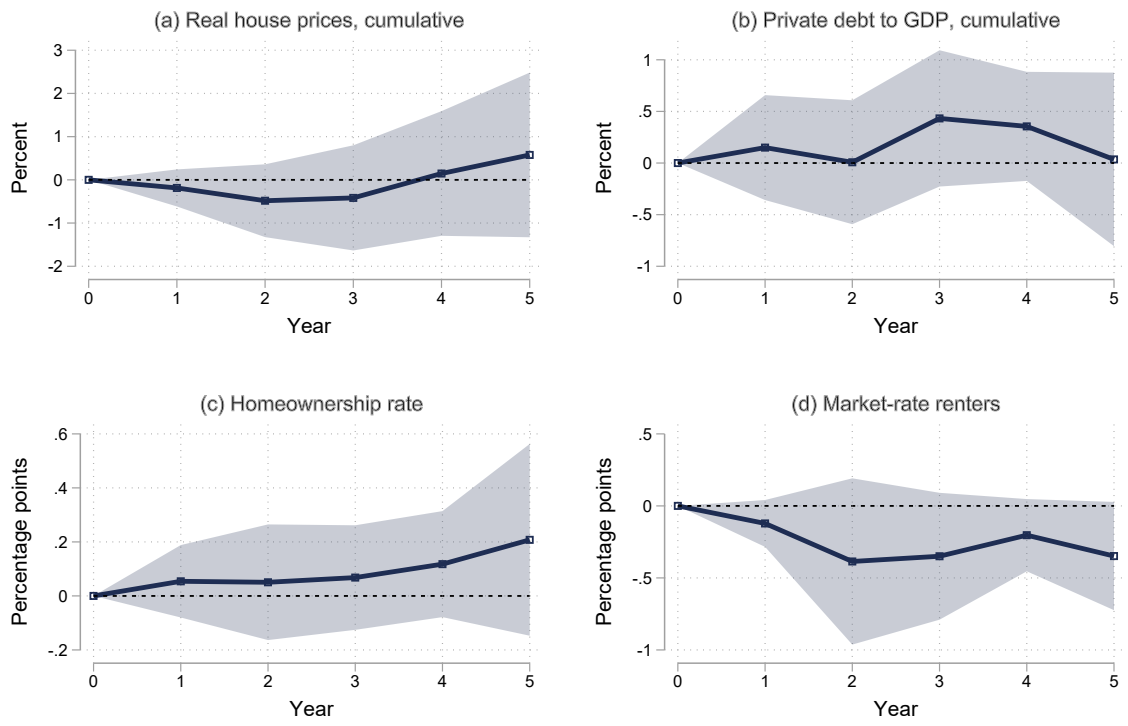


Figure 8: Prices, credit, ownership and market renting after a macroprudential tightening

Notes: Cumulative responses to a one-unit change in the macroprudential index with the Eurostat control set, 28 countries. Panels (a) and (b): percent change in real house prices and in private sector debt to GDP. Panels (c) and (d): change in percentage points in the aggregate homeownership rate and in the share of households renting at market rates. 95 percent confidence intervals in shaded areas, clustered by country.

The top row repeats the first-stage responses of house prices and private debt with controls; the bottom row shows the response of the aggregate homeownership rate and of the share of households renting at market rates, both with the full-coverage controls. Appendix Table B19 and Figure A13 report all rental margins, in aggregate and for young households.

The share of households renting at market rates does not rise. The cumulative responses are −0.12, −0.39, −0.35, −0.20 and −0.35 percentage points; only the five-year coefficient is significant, at the 10 percent level. The share renting at reduced rates or rent-free moves in the opposite direction by similar amounts, so the estimates point to a small shift from market to reduced-rate renting rather than to more households renting from landlords at market rates. The standard errors for the rental shares are larger than for the homeownership rate, and

equivalence at $\delta = 0.5$ percentage points fails at three of five horizons for market-rate renting. Among young adult-only households the estimates have the same signs and similar magnitudes. The margin on which the integrated framework predicts the clearest response, an increase in market renting, moves in the other direction, and the estimates are imprecise.

9 Large borrower-based reforms

The narrative index records changes in the regulatory stance, and several of the reforms the micro literature studies barely register in it (Section 5). This section asks whether those reforms are visible in national aggregates. I examine the introduction of loan-to-value and loan-to-income limits in the Netherlands in 2011 (Van Bekkum *et al.* , 2024), the United Kingdom in 2014 (Peydró *et al.* , 2024), Ireland in 2015 (Acharya *et al.* , 2022), Sweden in 2016 (Bäckman *et al.* , 2024) and Finland in 2016 (Eerola *et al.* , 2023), with the remaining 23 countries as controls. Each reform had documented effects on the borrowers it targeted. Constrained first-time buyers in Ireland lost 20 to 25 percent of their mortgage access, high loan-to-income lending in the United Kingdom fell by about 15 percent, and affected buyers in Finland entered ownership later. I report the Callaway & Sant'Anna (2020) estimator, which compares each treated country with never-treated and not-yet-treated countries only. Pre-reform coefficients are level differences relative to the year before the reform, so a positive pre-reform coefficient means that the outcome was higher two or three years before the reform than in the reference year, relative to controls. Because the sample ends in 2017, horizon three is identified from the Netherlands and the United Kingdom and horizon four from the Netherlands alone. The two-way fixed effects event study of the earlier version also uses already-treated countries as controls (Goodman-Bacon, 2021). Its estimates are in Appendix Table B27 and Appendix Figure A15.[5]

Figure 9 plots the event-time coefficients for real house prices, real household debt, the aggregate homeownership rate and the share of young households owning with a mortgage. Appendix

[5]The earlier version also included Norway, Switzerland and Portugal, which are outside the 28-country estimation sample. Appendix Figures A4 and A5 retain them.

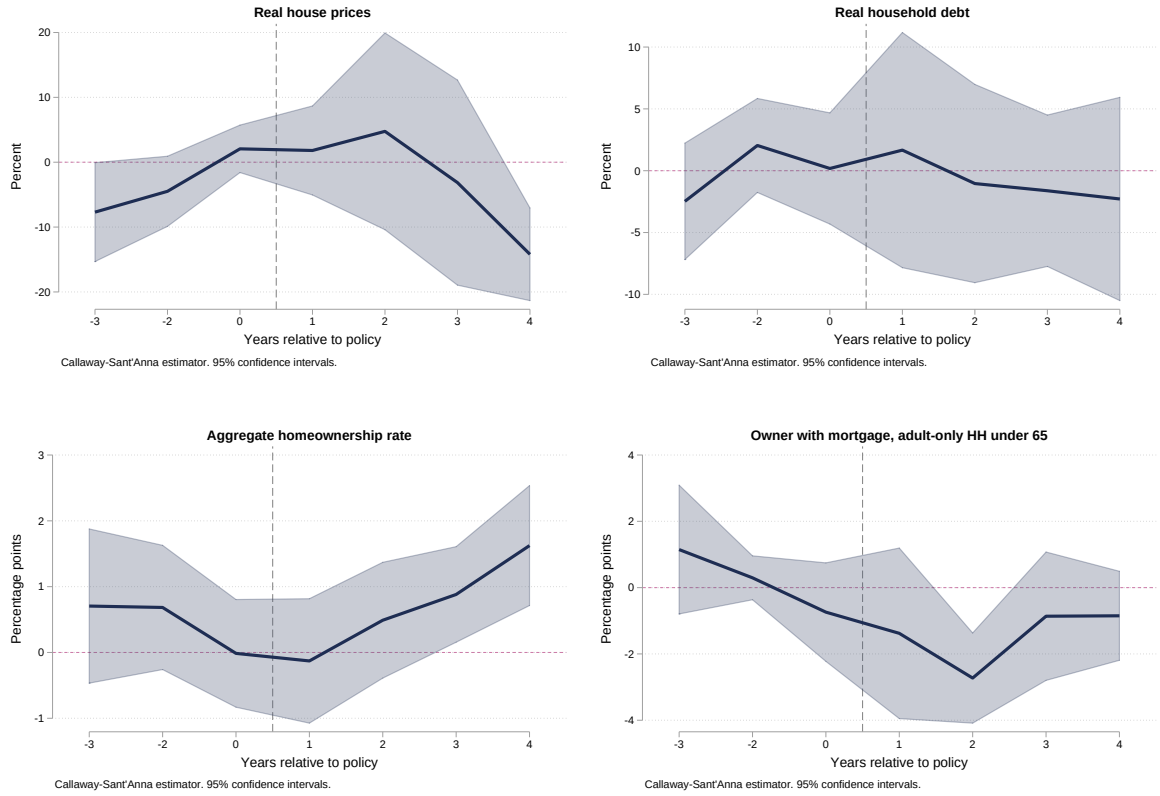


Figure 9: Event study around large borrower-based reforms: prices, credit and ownership

Notes: Callaway & Sant'Anna (2020) event-study coefficients for the log of the real house price index and the log of real household debt, both multiplied by 100 so that the units are percent (top row), and for the aggregate homeownership rate and the share of adult-only households under 65 who own with a mortgage, in percentage points (bottom row). Treated countries and years: the Netherlands (2011), the UK (2014), Ireland (2015), Sweden (2016) and Finland (2016); the other 23 countries and not-yet-treated countries serve as controls. The year before the reform is the reference period, and pre-reform coefficients are level differences relative to that year. Because the sample ends in 2017, horizon three is identified from the Netherlands and the UK and horizon four from the Netherlands alone. 95 percent confidence intervals in shaded areas, clustered by country. Averages over the pre- and post-reform windows are in Appendix Table B27; Appendix Figure A14 adds young households and market-rate renters.

Table B27 reports pre- and post-reform averages, and Appendix Table B26 reform-specific estimates, leave-one-out estimates and permutation inference. Credit and prices come first, because they determine how the ownership results can be read.

The reforms leave no common trace in aggregate credit or house prices. Real household debt moves by less than 2.5 percent at every horizon. The post-reform average is −0.6 percent with a standard error of 3.0, and the reform-specific estimates range from −11 percent for Ireland to +6 percent for the United Kingdom. Real house prices had risen by six to eight percent relative

to controls over the three years before the reforms, and the average pre-reform coefficient is −6.4 percent with a standard error of 3.3. After the reforms the relative rise stops. The coefficients stay within five percent of the reference year at horizons zero to three, none of them significant, and the post-reform average is −1.7 percent with a standard error of 4.0. The decline of 14 percent at horizon four comes from the Netherlands alone, where the 2011 cap was followed by the trough of the Dutch housing bust and by the 2013 requirement that new mortgages amortize to remain tax deductible. The reform-specific price estimates range from −10 percent for the Netherlands to +12 percent for Ireland.

Two readings of the missing first stage are possible, and they are not exclusive. The first is statistical. Household debt is a stock, and lending for house purchase adds perhaps 5 to 10 percent to it each year. A cut of 20 percent in lending to constrained first-time buyers, who account for a fraction of that flow, lowers the stock by roughly one to three percent after four years, inside confidence intervals of about eight percent on either side. The second reading is that equilibrium adjustments undo part of the effect measured on treated borrowers. The micro studies compare exposed with unexposed borrowers, lenders or regions within a country. Any response common to the whole country, including the response of prices and of lending to unexposed borrowers, is differenced out by construction, so these estimates are relative effects and are silent on the aggregate.The same studies record the adjustments that shrink the aggregate footprint. In Ireland, lenders reallocated mortgage credit toward higher-income borrowers and toward regions with slower house price growth (Acharya *et al.* , 2022). In Finland, affected buyers entered ownership later rather than not at all (Eerola *et al.* , 2023). In the segmented framework of Section 2.1, whatever demand shift remains is absorbed by prices rather than by the number of owners. These adjustments are part of how the policy works, and each drives a wedge between the effect on treated borrowers and the effect on national aggregates. The cross-country literature does find that borrower-based measures slow credit and house price growth (Kuttner & Shim, 2016; Cerutti *et al.* , 2017; Alam *et al.* , 2019), so aggregate effects are not absent. They are smaller and slower than the exposure of treated borrowers suggests, and with five reforms and annual national data they cannot be separated

from zero. The five reforms therefore do not supply the verified first stage that the average index lacks.

There is no common decline in the aggregate homeownership rate. The rate in treated countries had fallen by about 0.7 percentage points relative to controls between three years before the reforms and the year before them, and it rose after them. The average pre-reform coefficient is 0.78 percentage points with a clustered standard error of 0.65, and the post-reform average is 0.57 with a standard error of 0.33. Dropping one reform at a time gives post-reform averages between 0.04 and 0.78 percentage points. The reform-specific estimates are positive for the Netherlands and Ireland, close to zero for the United Kingdom and about -1 percentage point for Sweden and Finland, and the permutation p-value from reassigning the five treatment dates to random countries is 0.55.

The share of young households owning with a mortgage is lower after the reforms. The post-reform average is -1.31 percentage points with a standard error of 0.53, and between -0.9 and -1.7 when one reform is dropped. This is the one margin in the paper with a negative coefficient of conventional significance, and the margin on which the integrated framework predicts one. It does not survive closer inspection. The share had already fallen by about 1.1 percentage points relative to controls between three years before the reforms and the year before them, and the average pre-reform coefficient of 1.36 percentage points has a standard error of 0.73. The post-reform coefficients of -0.7, -1.4 and -2.7 percentage points at horizons zero to two continue that decline, and the coefficients of -0.9 at horizons three and four, from the Netherlands and the United Kingdom only, are smaller than the pre-reform decline. There is no break in the path of the young mortgagor share at the reforms. The permutation p-value is 0.28, and the reform-specific estimates range from 0.13 for the United Kingdom to -4.8 for Sweden. The homeownership rate of young households as a whole does not fall (post-reform average 0.30, standard error 0.75), and the share renting at market rates falls by 1.19 percentage points (standard error 0.46), the opposite of what elastic landlord conversion predicts. The two-way fixed effects estimates of -1.76 and -4.41 percentage points are two to three times larger, and Appendix Figure A15 shows that the difference comes from comparisons in which earlier-treated

countries serve as controls for later-treated ones.

Taken together, five well-documented borrower-based reforms were not followed by a common decline in aggregate credit, house prices or homeownership, and the one negative estimate, for young mortgagors, continues a relative decline that predates the reforms. The pre-reform paths show that the reforms came where house prices were rising and mortgaged ownership among the young was already falling relative to the rest of Europe. This is the endogeneity that the narrative index and the lagged controls in Section 5 address, and it means the comparison with never-treated countries is not clean. With five treated countries and heterogeneous institutions, the event studies are suggestive. What they do establish is a gap between the borrower-level and the aggregate evidence on the same reforms. The mechanisms behind that gap, reallocation by lenders, delayed rather than prevented entry, price adjustment and the dilution of flows in stocks, are the same ones that would keep the aggregate homeownership rate from responding to the average tightening. Section 10 makes the last of them explicit.

10 How large should the aggregate effect be?

A natural concern with the null results is that the aggregate homeownership rate may be too inertial to detect even substantial effects of macroprudential policies on the marginal buyer. The aggregate rate is a stock dominated by incumbent homeowners, while macroprudential policies primarily affect new mortgages. This section derives a simple stock-flow benchmark for the implied aggregate effect and compares it to the equivalence bounds estimated above and to the predictions of structural models. The benchmark is also the paper's account of why borrower-level effects need not appear in the stock. The aggregate rate records the cumulated flow of first-time buyers net of exits, so a policy that delays entry, or excludes a minority of entrants for a few years, leaves it almost unchanged.

The aggregate homeownership rate evolves according to a simple accounting identity. Let H_t denote the number of owner households and N_t the total number of households. Ignoring

household formation, the change in the homeownership rate is approximately

$$\Delta h_t \approx \frac{\text{FTB}_t - \text{Exits}_t}{N_t}, \tag{2}$$

where FTB_t denotes first-time entrants to homeownership and Exits_t denotes transitions from owning to renting. Repeat buyers do not enter the identity because they were already homeowners. A macroprudential tightening that reduces first-time buyer flows by a fraction α relative to baseline implies an aggregate effect of

$$|\Delta h_h| \approx h \cdot \alpha \cdot \phi, \tag{3}$$

where $\phi \equiv \text{FTB}_t/N_t$ is the baseline first-time buyer share of households and h is the horizon in years. Equation (3) assumes full persistence: excluded buyers never enter ownership. With partial recovery, where excluded buyers eventually enter after a delay, the implied effect is proportionally smaller. The micro evidence in Eerola *et al.* (2023) for Finland is consistent with partial recovery, as macroprudential rules delayed entry into homeownership rather than eliminating it.

Calibrating ϕ requires data on first-time buyer flows. Across European countries, ϕ ranges from approximately 0.8% to 1.8% per year, with a representative value near 1.3%.[6] For the policy-induced reduction α, I draw on the micro literature: Acharya *et al.* (2022) document that credit-constrained first-time buyers lost roughly 20–25% of mortgage access following the 2015 Irish LTV/LTI rules; Peydró *et al.* (2024) find the 2014 UK LTI cap reduced high-LTI mortgages by approximately 15%; and Bolliger *et al.* (2025) report transaction declines of more than 20% following the 2012 Swiss LTV tightening.

Table B6 reports implied magnitudes of $|\Delta h_h|$ under combinations of $\alpha \in \{10\%, 20\%, 30\%\}$ and horizons $h \in \{1, 3, 5\}$ years, holding $\phi = 1.3\%$ fixed. Two conclusions follow.

The benchmark is an illustrative calculation, not an accounting identity that converts the esti-

[6]Sources: UK Finance for the UK; Central Bank of Ireland for Ireland; Statistics Netherlands for the Netherlands. A first-time buyer share of 1.3% implies that the average household enters homeownership roughly once every 75 household-years, consistent with a typical first-purchase age of 30–35 in an adult life of 50–60 years.

mates into excluded buyers. It ignores changes in the number of households. Since the homeownership rate is a ratio, delayed household formation, for example young adults who continue to live with their parents because they cannot buy, lowers the denominator and can leave the measured rate unchanged while reducing access to independent ownership. It also treats the reduction in first-time-buyer flows as uniform across countries and years. With these limits in mind, two conclusions follow.

First, under full persistence and the largest empirically documented values of α, the implied five-year effect on the aggregate homeownership rate is 1.0–2.0 percentage points. This is comparable to the steady-state predictions of Kaplan *et al.* (2020) and Garriga & Hedlund (2020) of 2.8–3 percentage points, and lies outside the equivalence bounds estimated in Section 6, which are 0.5 percentage points for the cumulative response over four years and, in the specification without controls, over five. Inverting Equation (3) at $h = 5$, a cumulative margin of 0.5 percentage points corresponds to a permanent exclusion of $\alpha = 7.7$ percent of first-time buyers per unit of the index, or about 10 percent for the median tightening of 1.3 units (Appendix Table B11). Either the typical recorded tightening excludes far fewer than 20 percent of first-time buyers, excluded buyers eventually enter ownership, or the structural models overstate the aggregate response to the borrowing limits they study.

Second, for moderate values of α around 15% combined with partial recovery in first-time buyer flows over the five-year horizon, the implied effect on the aggregate rate falls below the 0.5 percentage point margin. The aggregate homeownership rate, on its own, cannot detect effects of macroprudential policies on the marginal buyer if those effects are temporary or offset by delayed entry. The evidence from Sections 7 and 9 is relevant here. The share of young households owning with a mortgage does not fall after the average change in the index; after the five large reforms it is about 1.3 percentage points lower than in control countries, from a base of about 40 percent of young households in those countries, but it had already fallen by a similar amount relative to controls before the reforms, and the permutation p-value is 0.28. A permanent reduction of 20 percent in first-time buyer flows would lower the young mortgagor share by several percentage points over five years, since young households account

for most first-time buyers. The evidence on this margin is therefore consistent with the aggregate bound: exclusion of the size documented in the micro literature for individual reforms has not persisted on the scale required to move the stock. The results are silent on the marginal buyer's welfare, on temporary delays in entry, and on household formation, all of which require micro evidence to address.

11 Robustness

Pre-trends. If changes in the index respond to past movements in homeownership, the local projections would pick up reversion rather than a policy effect. Appendix Figure A9 extends the cumulative response to three years before the change in the index, and Appendix Table B14 reports regressions of the annual changes in homeownership one to three years before the index change on the index change. Without controls, the three coefficients are 0.04, −0.02 and 0.00 percentage points, none significant, and a joint test that the lagged changes predict the index change has a p-value of 0.67. With the full-coverage controls, the changes one and two years before the policy are −0.12 and −0.13 percentage points, each significant at the 10 percent level, with a joint p-value of 0.22. There is weak evidence that changes in the index follow small declines in homeownership once macroeconomic conditions are held fixed. Reversion after such declines would bias the post-policy coefficients upward, which is a reason to give more weight to the specification without controls, where there is no pre-trend and the estimates are closer to zero.

Tightenings and loosenings. The index treats a loosening as the negative of a tightening. Appendix Table B24 enters the positive and negative parts of the index change separately. The tightening coefficients are 0.08, 0.04, −0.02, −0.01 and −0.03 percentage points without controls and 0.06 to 0.21 with controls, with standard errors of 0.07 to 0.17, so the results above are results about tightenings. The loosening coefficients are based on 17 country-years and have standard errors of 0.4 to 1.3 percentage points; they range from −2.1 to 1.0 across horizons and specifications. For lending standard restrictions there are two loosening country-years, and the

coefficients are not interpretable. The data do not identify the effect of loosening.

Inference. With 28 clusters, asymptotic clustered standard errors may be unreliable. Appendix Table B25 reports, for the aggregate cumulative response, p-values and confidence intervals from a wild cluster bootstrap with Webb weights (Roodman *et al.* , 2019), Driscoll-Kraay standard errors that allow for cross-sectional dependence, the range of coefficients obtained by dropping each country in turn, and population-weighted estimates. The wild cluster bootstrap p-values are within 0.03 of the clustered p-values at every horizon, and the bootstrap 95 percent confidence intervals for the specification with controls are $[-0.07, 0.21]$ at one year and $[-0.18, 0.66]$ at five years. Driscoll-Kraay standard errors are similar to or smaller than the clustered ones. Dropping any single country moves the five-year coefficient by at most 0.1 percentage points in the specification with controls and 0.05 without. Weighting country-years by population lowers the coefficients in the specification without controls, to -0.24 percentage points with a standard error of 0.28 at five years, and leaves the specification with controls close to unweighted, at 0.26 with a standard error of 0.13. None of these variants shows a decline in the homeownership rate at conventional significance levels.

Randomization inference. I reshuffle the index change across countries within each year 2,000 times and re-estimate the cumulative local projection without controls each time. Appendix Table B15 and Figure A10 report the results. The randomization p-values are 0.51, 0.95, 0.68, 0.78 and 0.57 at horizons one to five, and the p-value for the largest absolute coefficient across horizons is 0.81. The 97.5th percentile of the permuted coefficients rises from 0.13 percentage points at one year to 0.28 at five years, which indicates the size of effect the design could have detected at each horizon.

Alternative policy database. Appendix Figure A3 shows that the results for all groups are similar when the IMF's integrated Macroprudential Policy (iMaPP) database, originally constructed by Alam *et al.* (2019), replaces MaPPED.

Control sets and house price source. Table B10 compares the BIS and Eurostat control sets. Neither shows a decline; the BIS set, on 18 countries, shows positive coefficients at four and five years. Real house price growth from Eurostat and from the BIS have a correlation of 0.99 where both are available, and the results are unchanged when the BIS index is used throughout.

12 Conclusion

Recorded changes in macroprudential regulation are not followed by detectable declines in homeownership rates in Europe. Across 28 countries and fifteen years, changes in a narrative index of non-counter-cyclical macroprudential actions leave the aggregate homeownership rate within half a percentage point of its previous level over horizons of up to four years, and the estimates are precise enough to reject persistent aggregate effects of that size. The same holds for households with and without a mortgage, for young households, for the share of young households owning with a mortgage, and for the share of households renting at market rates. The estimates for low-income households are less precise. Around the large borrower-based reforms in the Netherlands, the United Kingdom, Ireland, Sweden and Finland, there is no common decline in the aggregate rate, and aggregate credit and house prices show no common movement either. The share of young households with a mortgage is about one percentage point lower after the reforms than in control countries, but it had already fallen relative to controls by a similar amount in the years before, the estimate is not significant under permutation inference, and the reform-specific estimates differ in sign.

Subject to the identification and measurement assumptions, the evidence rules out sufficiently large and persistent aggregate effects, while remaining consistent with temporary delays in entry and with effects concentrated on marginal buyers that do not accumulate in the stock. The results are consistent with the segmented-market view of Greenwald & Guren (2025), in which credit shocks move house prices and leave the homeownership rate unchanged because the supply of owner-occupied housing relative to rental housing is inelastic, and they are hard to reconcile with the aggregate magnitudes of 1.8 to 3 percentage points predicted by the central

calibrations of models with elastic landlord conversion. They do not establish segmentation, because house prices do not respond in these data either. The index has a weak first stage on credit and house prices, a one-unit change in the index is a count of actions rather than a change in a borrowing limit of known size, and the large reforms do not produce a common movement in aggregate credit or prices in annual national data. The results are therefore also consistent with regulatory actions that had little effect on aggregate credit, and with borrower-level effects that were reallocated, delayed or absorbed by prices before they reached the stock of owners. Slow adjustment remains possible: five years may be too short for households delayed by a policy to appear in the stock, and the stock-flow benchmark shows that temporary delays would not be visible in the aggregate rate at all. Delayed household formation would not be visible either.

For modeling, the results suggest that the aggregate homeownership response to borrowing constraints in quantitative housing models is too large, and that the response should be concentrated in the timing of entry rather than in the stock of owners. Segmentation between rental and owner-occupied housing, whether through landlord technology, rental market regulation or the composition of the housing stock, is one way to obtain this. For policy, the results do not support the concern that macroprudential tightenings have reduced homeownership in aggregate. Over horizons of up to five years, the aggregate cost of recorded tightenings in terms of homeownership has been small. This does not mean that borrower-based measures have no distributional effects. The micro literature documents that they change who obtains credit, and in the five reform countries the share of young households owning with a mortgage was falling relative to the rest of Europe before the reforms and continued to fall afterwards. What the aggregate evidence shows is that these effects do not accumulate in the stock of owners, because within-country estimates difference out the aggregate response, lenders reallocate credit, excluded buyers enter later, and stocks dilute flows. The results do not show that relaxing borrower-based measures would leave homeownership unchanged, because loosenings are rare in the sample and their effects are not identified. They are silent on the welfare of the marginal buyer, on delays in entry, on household formation, and on other costs of macroprudential reg-

ulation, such as the reduction in household liquidity documented by Aastveit *et al.* (2023). Those questions require micro data, and the aggregate evidence here defines the magnitudes they should be reconciled with.

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Online Appendixes

A Online Appendix: Figures

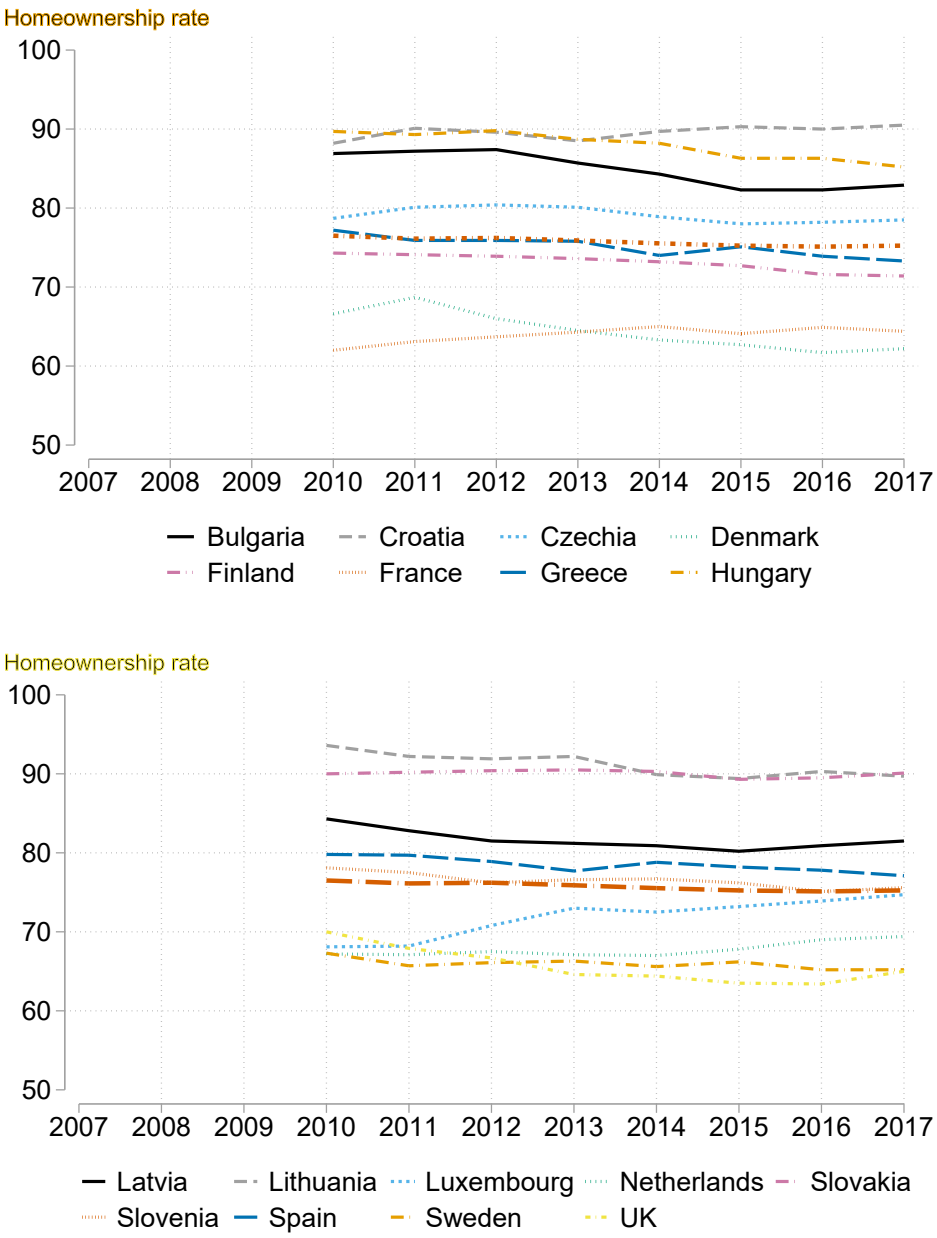


Figure A1: Homeownership rates across countries

Note: The figure shows the homeownership rate over time for all countries in the sample. The data source is EU-SILC. Panel a) plots results for Bulgaria, Croatia, Czechia, Denmark, Finland, France, Greece, and Hungary. Panel b) plots results for Latvia, Lithuania, Luxembourg, Netherlands, Slovakia, Slovenia, Spain, Sweden, and UK.

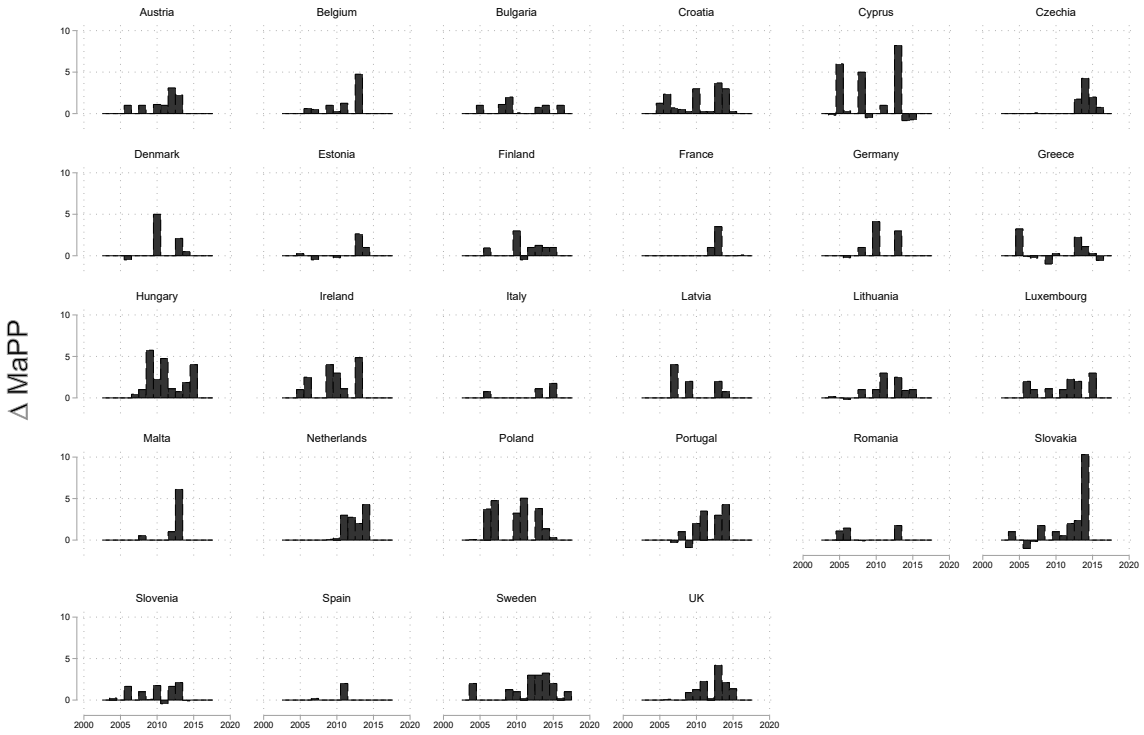


Figure A2: Macroprudential policy shocks by country

Note: The figure shows the macroprudential policy index over time for all countries in the sample. The data source is MaPPED database. Only policies included in the empirical analysis are included. This excludes policies with a countercyclical design.

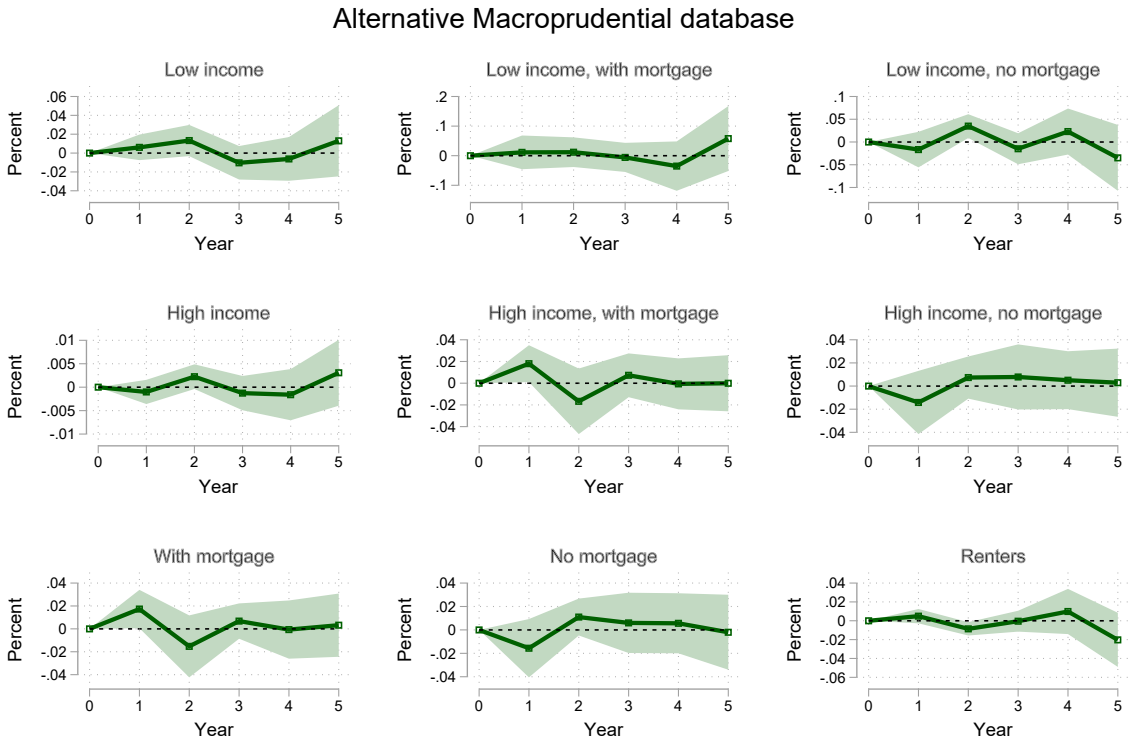


Figure A3: Linear projections with alternative macroprudential policy index

Note: The figure shows the impulse response functions for to an alternative macroprudential policy database over a five-year horizon, estimated using equation (1). The dependent variable is the homeownership rate, and the independent variable is the change in the cumulative sum of macroprudential policies from the iMaPP database. The red line represents the specification with control variables included, and the blue line represents the specification with control variables excluded. Control variables are all lagged one year, and include GDP growth rate per capita, inflation, the change in the house price index, the change in credit to the private non-financial sector relative to GDP, the change in the central bank interest rate, and the sum of macroprudential policies. 95 percent confidence intervals are marked in shaded areas.

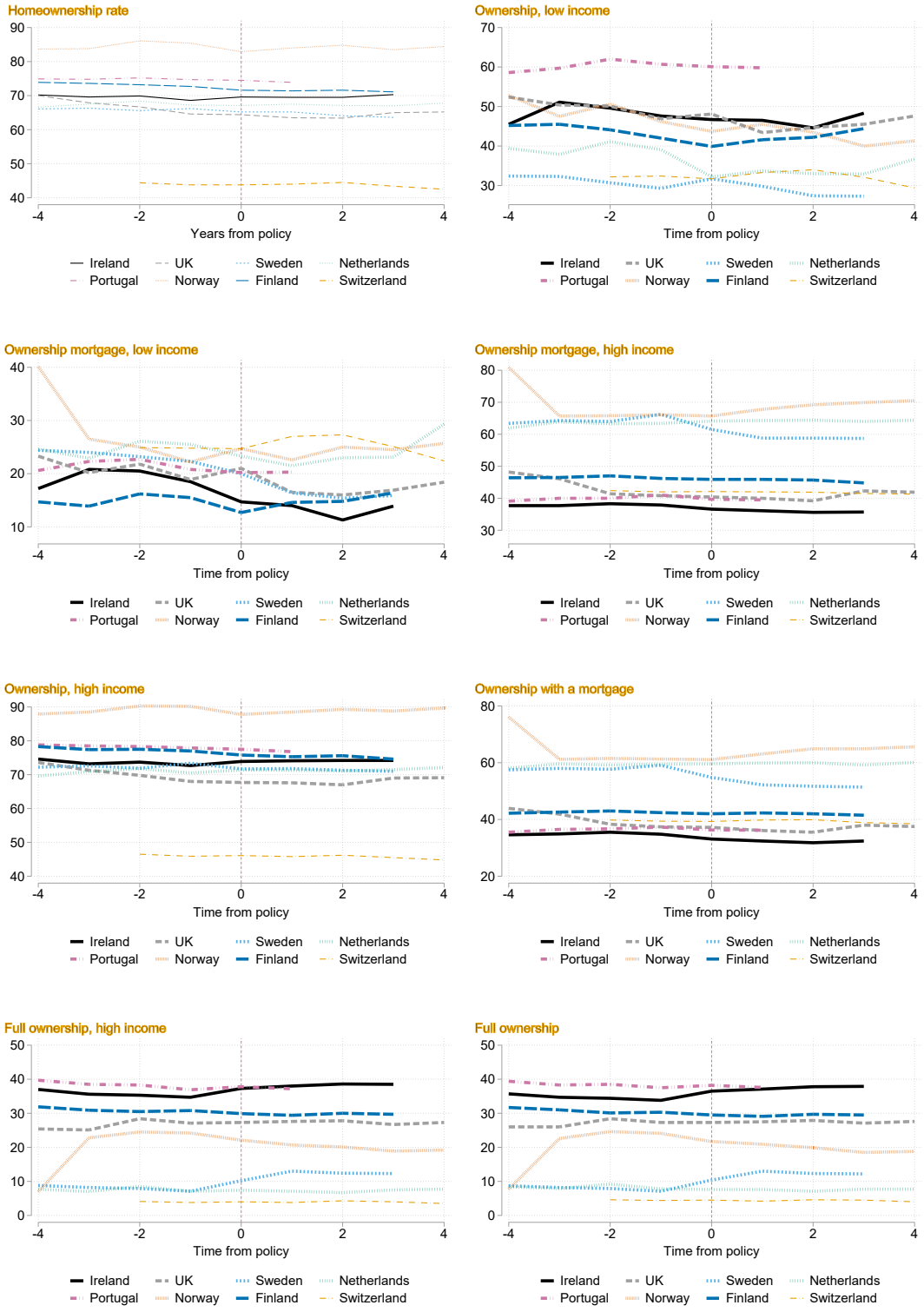


Figure A4: Homeownership rates around large macroprudential policy shocks

Note: The figure plots different homeownership rates around large macroprudential policy shocks. The countries (year of policy) are Ireland (2015), the UK (2014), Sweden (2016), the Netherlands (2011), Portugal (2018), Norway (2010), Finland (2016) and Switzerland (2012). The data source is EU-SILC. Each panel represents a different ownership group, stated above the y-axis.

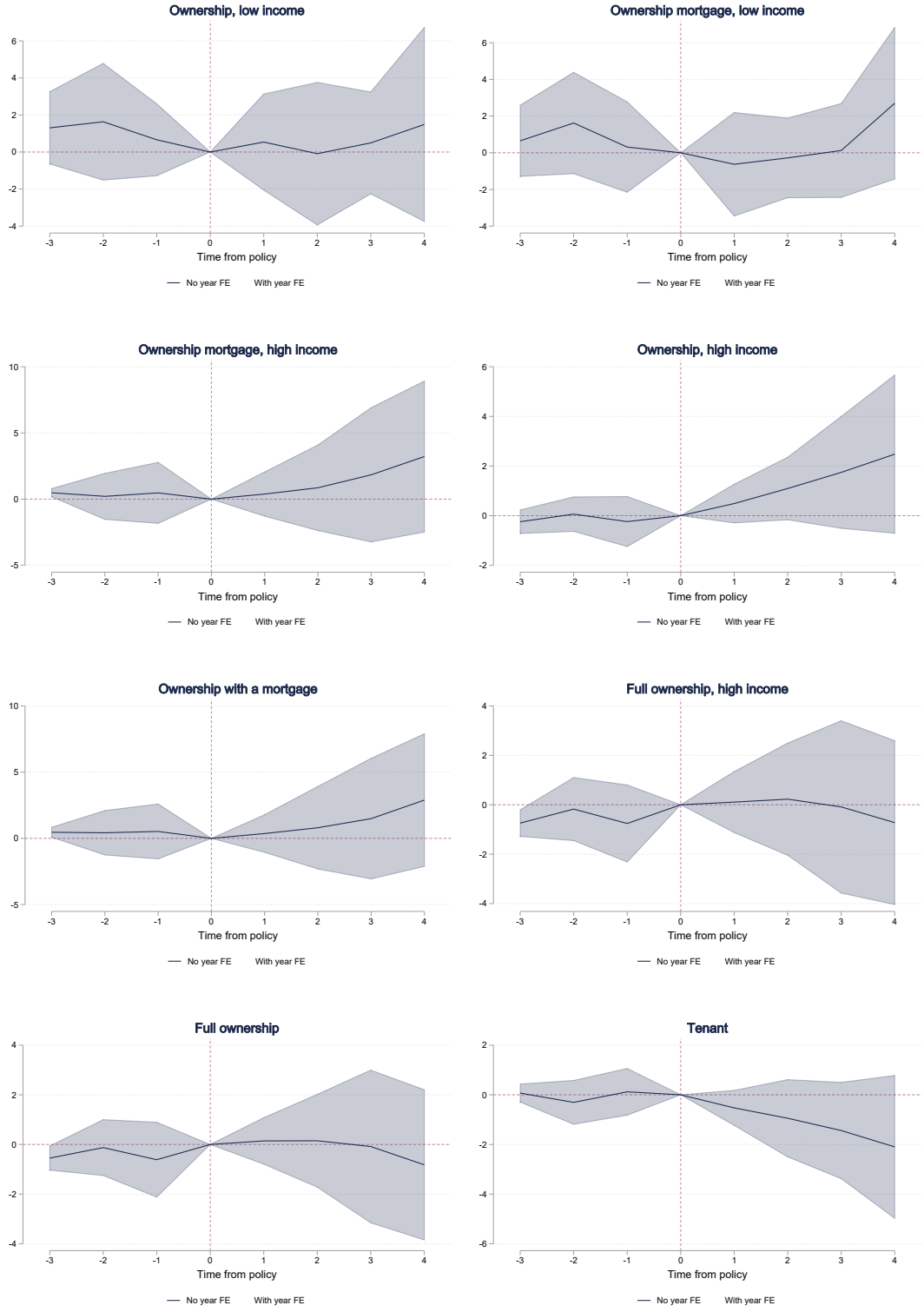


Figure A5: Coefficient estimates around large macroprudential policy shocks

Note: The figure plots the coefficients from a two-way fixed effects event study of each homeownership rate on country and year fixed effects and indicators for years relative to the reform, for different homeownership groups. The countries (year of policy) are Ireland (2015), the UK (2014), Sweden (2016), the Netherlands (2011), Portugal (2018), Norway (2010), Finland (2016) and Switzerland (2012). The time from event measures the time from the large macroprudential shock. Standard errors are clustered at the country level. 95 percent confidence intervals are marked in shaded areas.

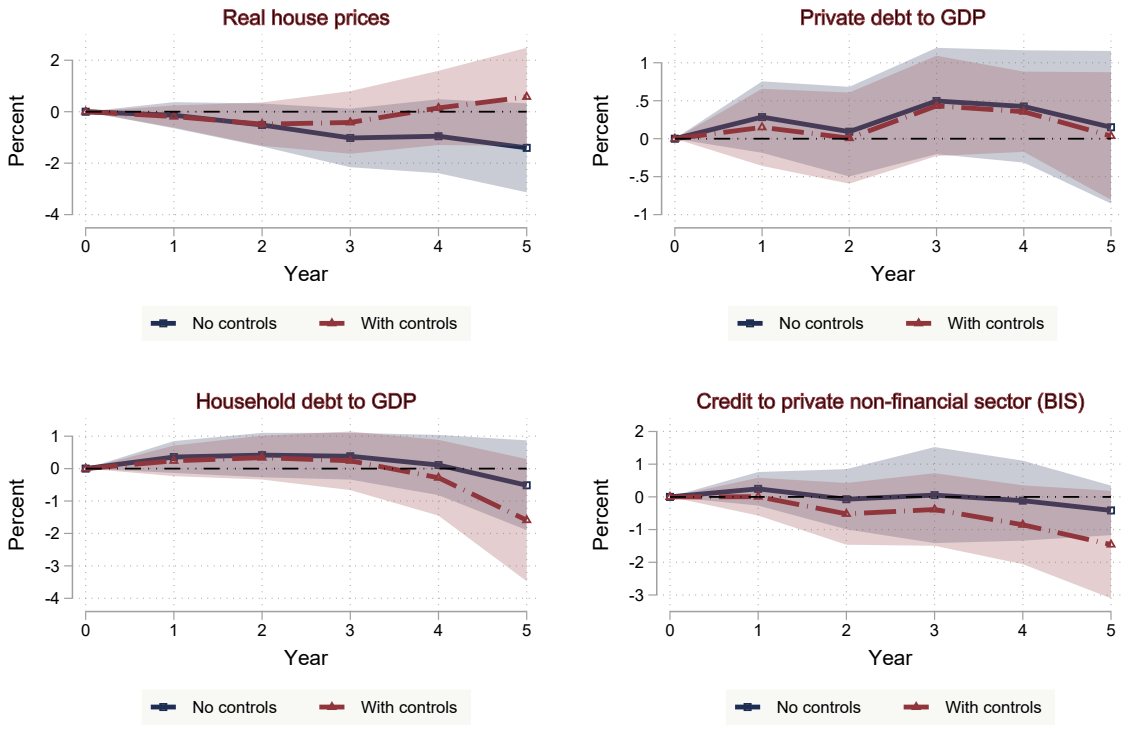


Figure A6: First-stage responses of house prices and credit to the macroprudential index

Note: Cumulative responses (percent) of real house prices, Eurostat private sector debt to GDP, Eurostat household debt to GDP and BIS credit to the private non-financial sector to GDP to a one-unit change in the macroprudential index. The blue line excludes and the red line includes control variables. The BIS series is available for 18 countries, the others for 28. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Table B12.

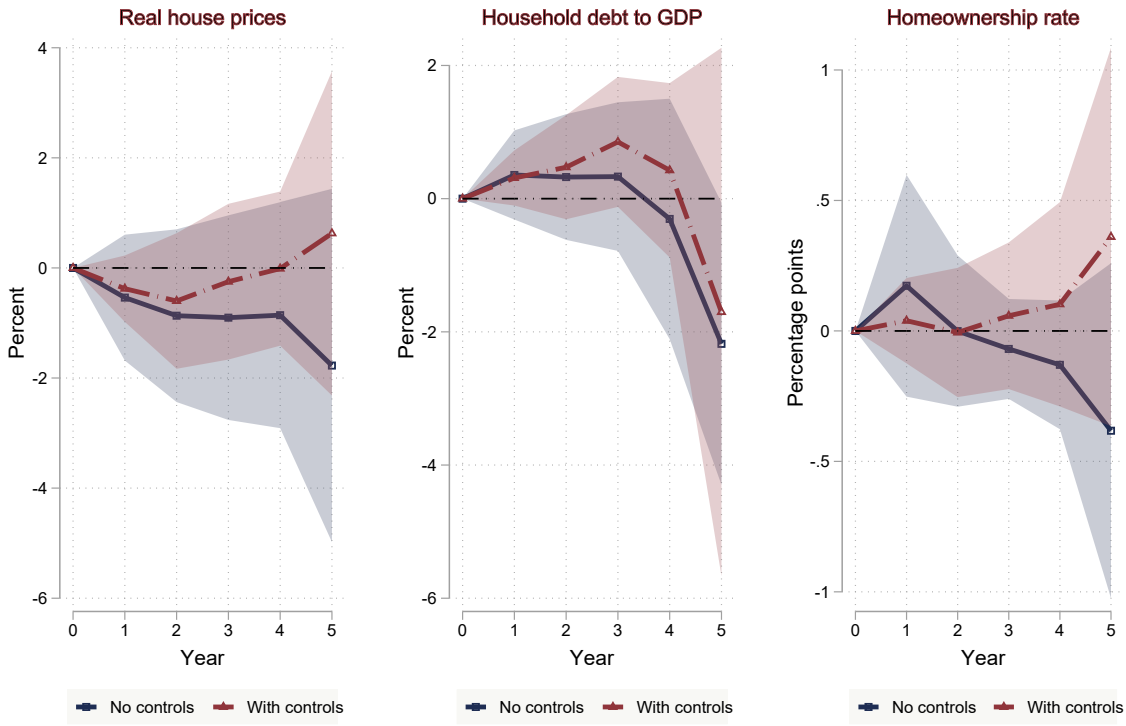


Figure A7: Responses to lending standard restrictions: house prices, household debt and homeownership

Note: Cumulative responses of real house prices and household debt to GDP (percent), and the response of the aggregate homeownership rate (percentage points), to a one-unit change in the index of lending standard restrictions. The blue line excludes and the red line includes control variables. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Table B13.

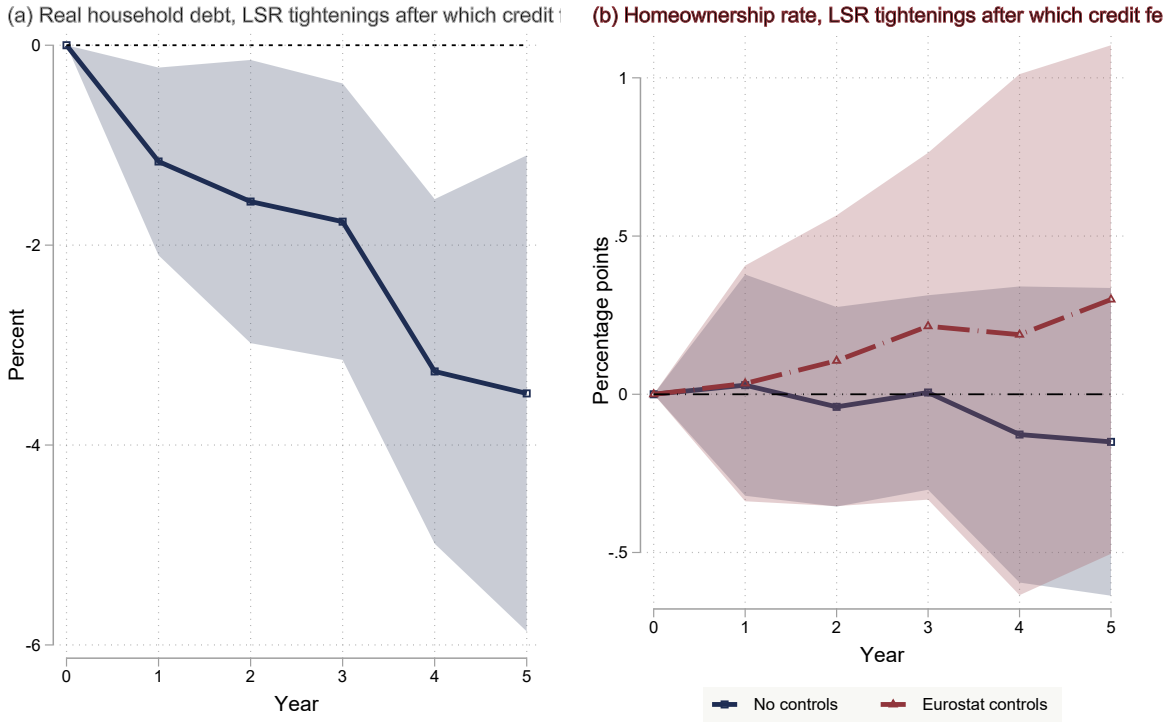


Figure A8: Lending standard restrictions followed by a fall in household credit: credit and homeownership responses

Note: Panel (a): cumulative percent response of real household debt to tightenings of lending standard restrictions after which real household debt fell between $t-1$ and $t+2$. Panel (b): response of the aggregate homeownership rate, in percentage points, to the same tightenings, without controls (blue) and with the Eurostat control set (red). Tightenings not followed by a fall in credit enter the same regressions as a separate variable. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Table B22.

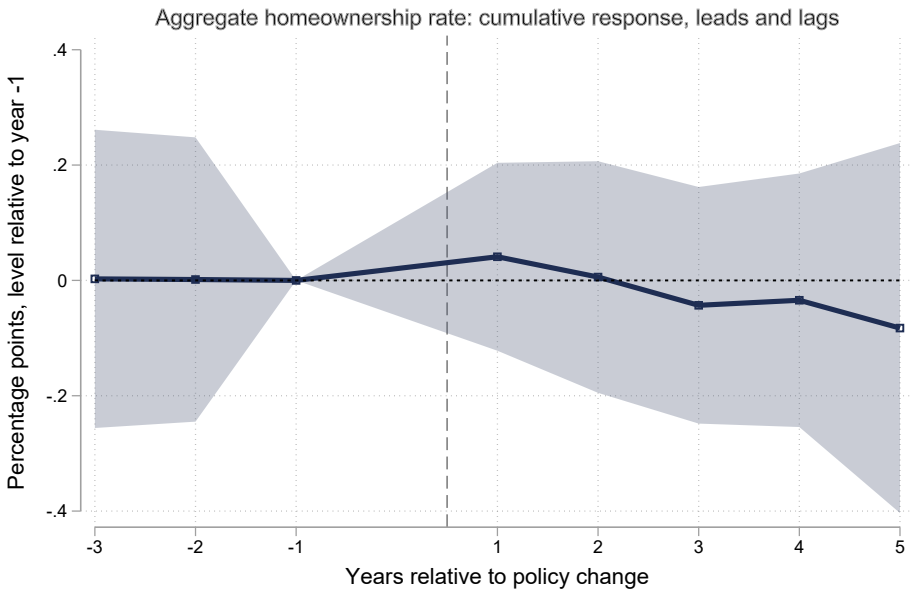


Figure A9: Changes in the homeownership rate before and after a change in the macroprudential index

Note: Cumulative change in the homeownership rate relative to year -1, the year before the index change, for years -3 to 5, from regressions on the change in the macroprudential index with country and year fixed effects and no controls. Year -1 is the reference period. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients for $h < 0$ are in Table B14.

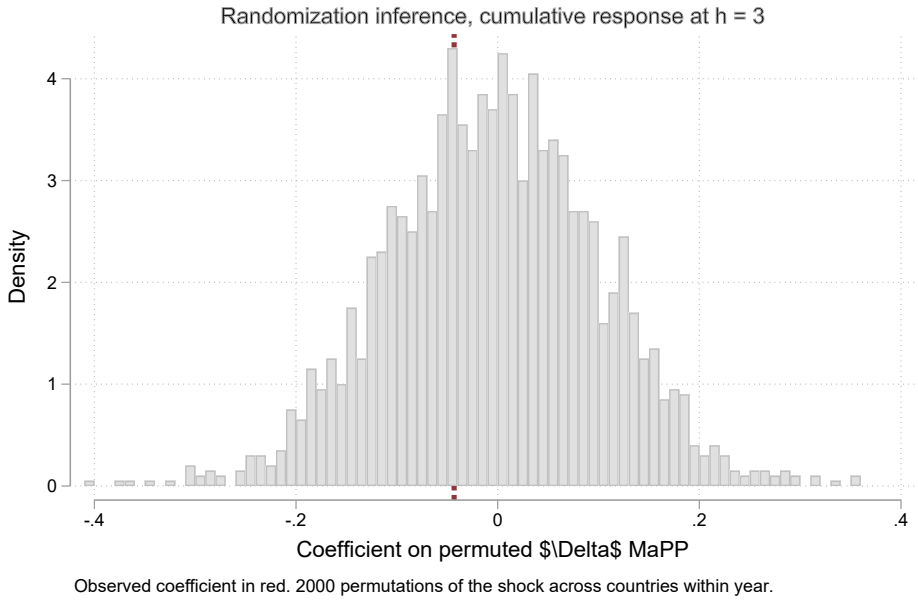


Figure A10: Randomization inference: distribution of placebo coefficients at $h = 3$

Note: Distribution of the cumulative-response coefficient on the macroprudential index change at horizon three when the index change is reshuffled across countries within each year (2,000 permutations). The red line is the observed coefficient. P-values for all horizons are in Table B15.

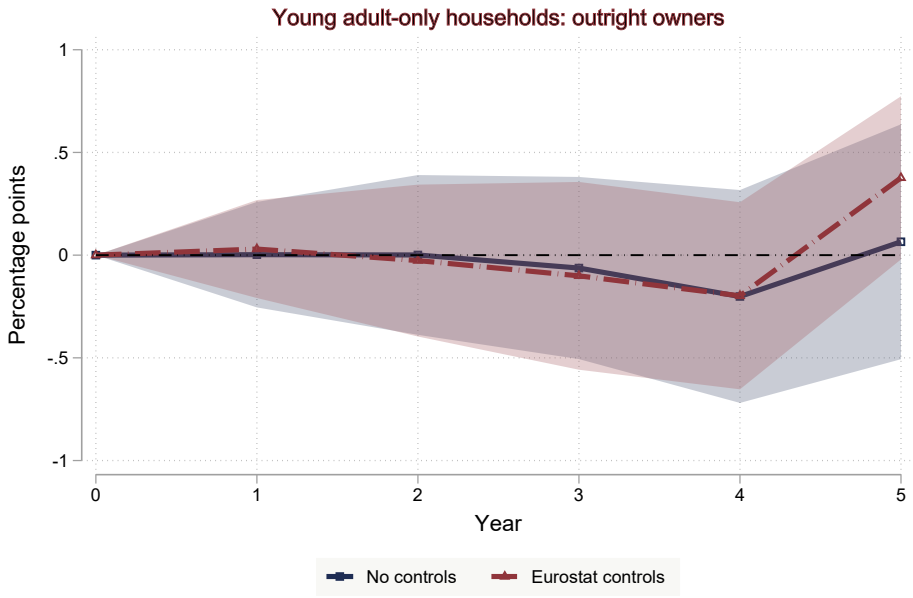


Figure A11: Young adult-only households: share owning outright

Note: Impulse responses, in percentage points, of the share of adult-only households under 65 who own their home without a mortgage to a one-unit change in the macroprudential index. The blue line includes country and year fixed effects only; the red line adds the Eurostat control set. 95 percent confidence intervals in shaded areas, clustered by country.

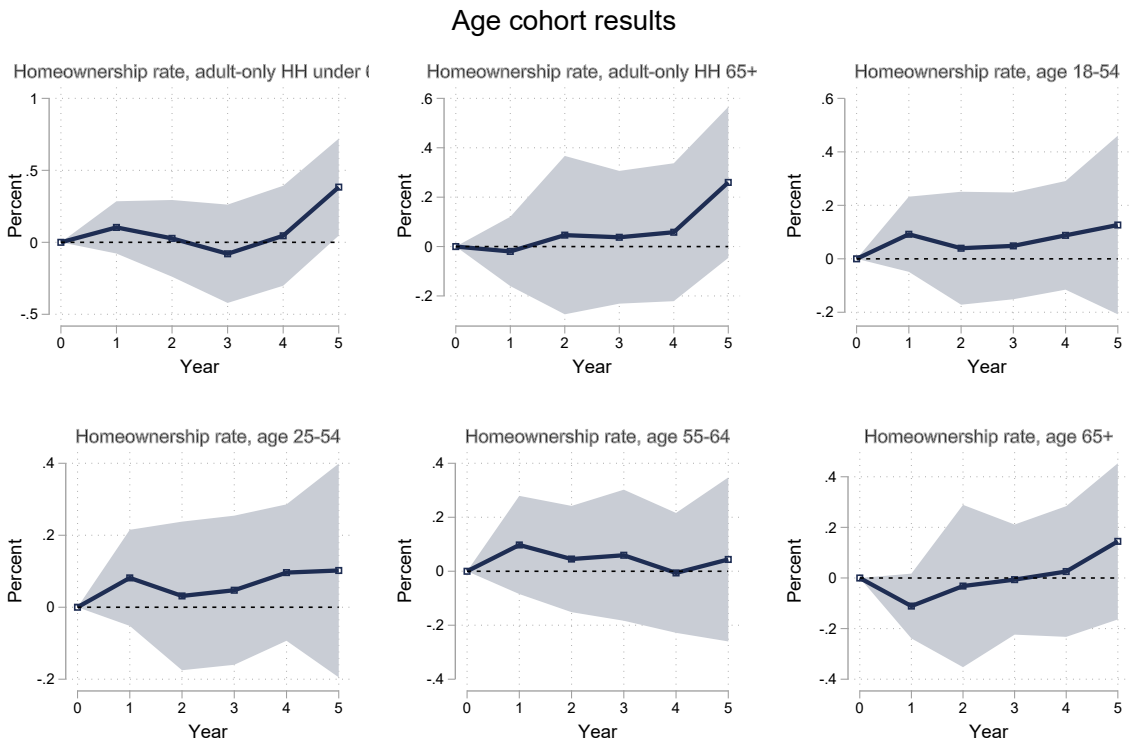


Figure A12: Linear projections of homeownership rates by age group

Note: Impulse responses of the homeownership rate of adult-only households under 65 and 65 and over, and of individuals aged 18–54, 25–54, 55–64 and 65 and over, to a one-unit change in the macroprudential index, cumulative responses with the Eurostat control set. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Table B17.

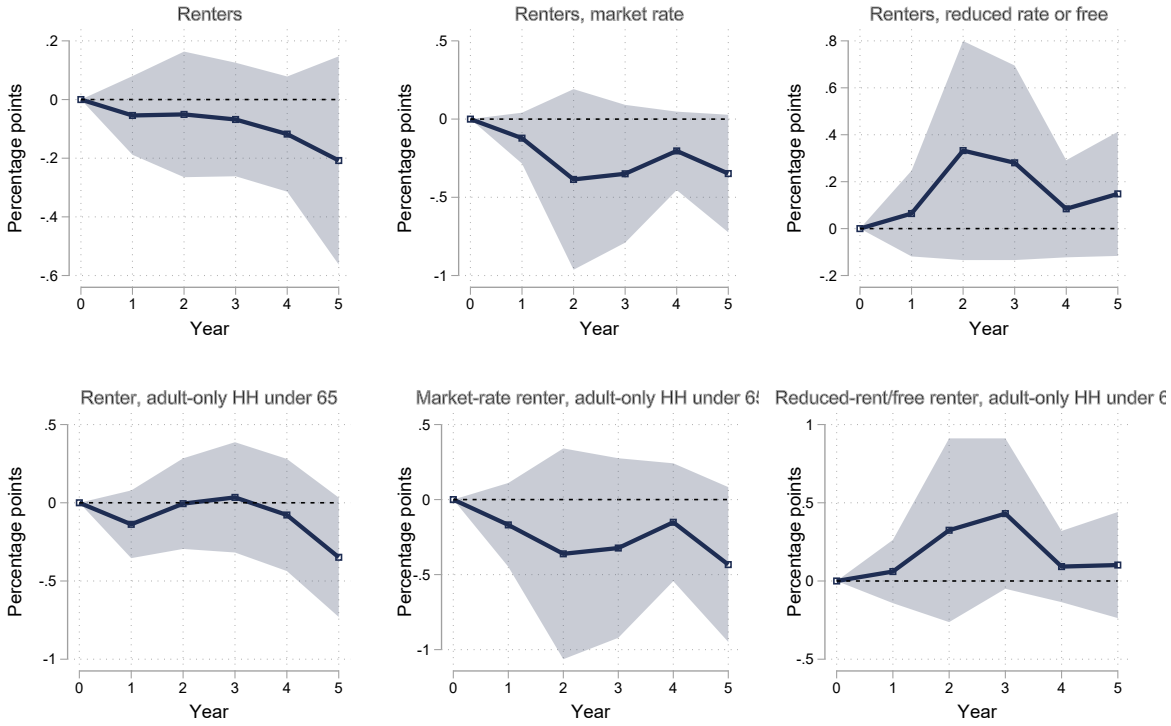


Figure A13: Linear projections of rental market shares

Note: Impulse responses of the share of households renting, renting at market rates and renting at reduced rates or rent-free, for all households (top row) and for adult-only households under 65 (bottom row), to a one-unit change in the macroprudential index, with the Eurostat control set. 95 percent confidence intervals in shaded areas, clustered by country. Coefficients are in Table B19.

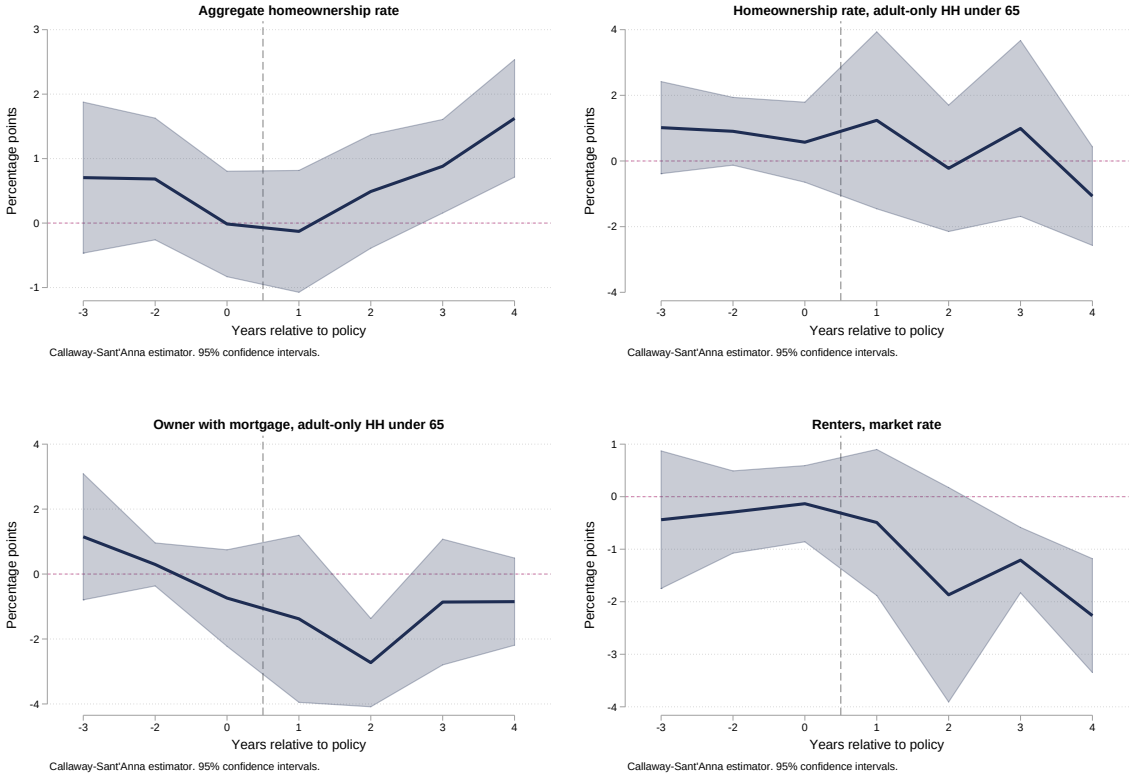


Figure A14: Staggered event study around large borrower-based policy shocks

Note: Callaway & Sant'Anna (2020) event-study coefficients for the aggregate homeownership rate, the homeownership rate of adult-only households under 65, the share of those households owning with a mortgage, and the share of all households renting at market rates. Treated countries and years: the Netherlands (2011), the UK (2014), Ireland (2015), Sweden (2016) and Finland (2016); never-treated and not-yet-treated countries serve as controls. The year before treatment is the reference period. 95 percent confidence intervals in shaded areas, clustered by country. Averages are in Table B27.

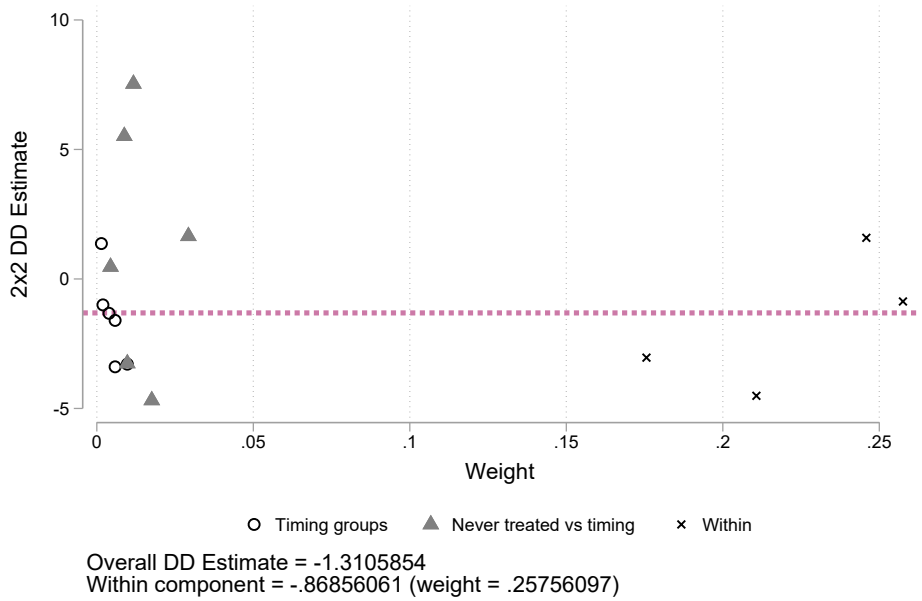


Figure A15: Goodman-Bacon decomposition of the two-way fixed effects estimate

Note: The two-way fixed effects estimates in Table B27 regress each outcome on country and year fixed effects and a post-reform indicator. With staggered adoption and effects that differ across countries or over time, this estimator also uses already-treated countries as controls for later-treated ones. The figure decomposes the coefficient for the aggregate home-ownership rate into two-by-two difference-in-differences comparisons, following Goodman-Bacon (2021), on the balanced sub-panel 2005–2017.

B Online Appendix: Tables

Table B1: Weighting Scheme for Different Macroprudential Policy Actions in Narrative Measure

Type of Policy Action	Weight	Strengthening / Loosening	Sign	Final Weight
Activation	1	Tightening	+	1
		Other/ambiguous impact		0
		Loosening	-	-1
Change in the Level	0.25	Tightening	+	0.25
		Other/ambiguous impact		0
		Loosening	-	-0.25
Change in the Scope	0.10	Tightening	+	0.10
		Other/ambiguous impact		0
		Loosening	-	-0.10
Maintaining the Existing Level and Scope	0.05	Tightening	+	0.05
		Other/ambiguous impact		0
		Loosening	-	-0.05

Notes: This table shows the weighting scheme used to construct the narrative measure of macroprudential policy actions. The final weight is calculated as the product of the base weight and the sign (+1 for tightening, -1 for loosening, 0 for ambiguous). Source: Meuleman & Vander Vennet (2020).

Table B2: Homeownership sample

			Homeownership rate			
Country	First year	Last year	First year	Last year	Average	Difference
Austria	2007	2017	59	55	57	-4
Belgium	2003	2017	73	72	72	-1
Bulgaria	2005	2017	85	83	85	-2
Croatia	2010	2017	88	90	90	2
Cyprus	2007	2017	74	71	73	-3
Czechia	2005	2017	74	78	77	4
Denmark	2003	2017	66	62	66	-4
Estonia	2006	2017	88	82	84	-6
Finland	2004	2017	71	71	73	0
France	2005	2017	62	64	63	2
Germany	2010	2017	53	51	52	-2
Greece	2007	2017	76	73	75	-3
Hungary	2005	2017	88	85	88	-3
Ireland	2003	2017	80	70	74	-10
Italy	2004	2017	73	72	73	-1
Latvia	2007	2017	86	82	83	-4
Lithuania	2005	2017	88	90	91	2
Luxembourg	2007	2017	74	75	72	1
Malta	2005	2017	80	81	80	1
Netherlands	2005	2017	64	69	67	5
Poland	2007	2017	62	84	78	22
Portugal	2004	2017	75	75	75	0
Romania	2007	2017	96	97	96	1
Slovakia	2005	2017	82	90	89	8
Slovenia	2005	2017	83	76	79	-7
Spain	2007	2017	81	77	79	-4
Sweden	2004	2017	67	65	67	-2
UK	2005	2017	70	65	68	-5

Notes: The table provides information about available homeownership data for each country in our sample. The first column lists the first year where homeownership data is available and the second column lists the last year when homeownership data is available. The final three columns provides data on the homeownership rate in the first year, in the last year and on average for all years. Germany (2006, 2007, 2008 and 2009), Greece (2004, 2005 and 2006) and Turkey (2010) are missing data for select years.

Table B3: The effect of macroprudential policies on the homeownership rate

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Panel A: No controls					
Δ MaPP	0.041	0.006	-0.043	-0.034	-0.083
	(0.079)	(0.098)	(0.100)	(0.107)	(0.156)
Observations	319	291	263	235	207
Economic significance	0.04	0.00	0.02	0.02	0.03
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Panel B: With controls					
Δ MaPP	0.054	0.051	0.068	0.118	0.208
	(0.065)	(0.104)	(0.094)	(0.096)	(0.173)
L.Δ GDP per capita	5.425**	10.379**	23.743***	18.538**	21.103**
	(2.358)	(3.784)	(6.140)	(7.818)	(7.843)
L.Inflation	13.458***	8.327	6.785	-6.372	-18.270
	(3.944)	(7.201)	(9.508)	(11.953)	(14.439)
L.Δ House prices	-0.177	1.163	1.212	2.537	1.504
	(1.515)	(2.258)	(2.691)	(3.305)	(3.317)
L.Δ Private debt to GDP	0.535	2.506	3.177	4.624*	7.553**
	(1.512)	(1.729)	(1.941)	(2.633)	(3.200)
L.Δ 3-month money market rate	0.042	0.192*	0.236*	0.251**	0.232**
	(0.084)	(0.096)	(0.117)	(0.110)	(0.093)
L.Macroprudential policy index (MaPP)	0.020	0.032	0.051	0.082	0.194
	(0.033)	(0.061)	(0.101)	(0.194)	(0.258)
Observations	272	245	218	190	162
Economic significance	0.06	0.04	0.04	0.05	0.08
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No

Notes: The dependent variable is the cumulative change in the homeownership rate from the year before the index change, and the independent variable is the change in the macroprudential index. Panel B uses the Eurostat control set, which covers all 28 countries; "L" denotes lagged values. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B4: The effect of macroprudential policies on the homeownership rate for different groups

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Dep. var.: Ownership, low income					
Δ MaPP	-0.184	0.155	0.059	0.020	-0.015
	(0.180)	(0.213)	(0.204)	(0.272)	(0.257)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.076	-0.044	0.009	0.086	0.240
Equivalence (CI fully inside margin):	Yes	No	Yes	Yes	Yes
Dep. var.: Ownership mortgage, low income					
Δ MaPP	-0.044	0.122	0.042	0.084	-0.432*
	(0.136)	(0.148)	(0.115)	(0.258)	(0.230)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.092	0.020	0.121	0.195	0.268
Equivalence (CI fully inside margin):	Yes	Yes	Yes	No	No
Dep. var.: Full ownership, low income					
Δ MaPP	-0.141	0.031	0.011	-0.059	0.419
	(0.187)	(0.224)	(0.218)	(0.330)	(0.284)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	0.009	0.104	0.141	0.201	0.291
Equivalence (CI fully inside margin):	Yes	Yes	Yes	No	No
Dep. var.: Ownership, high income					
Δ MaPP	0.089	0.017	0.076	0.157	0.299
	(0.065)	(0.107)	(0.103)	(0.109)	(0.203)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	0.018	0.203	0.362	0.455	0.608
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No
Dep. var.: Ownership mortgage, high income					
Δ MaPP	0.003	0.053	0.003	0.130	-0.056
	(0.094)	(0.119)	(0.164)	(0.180)	(0.245)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.009	0.083	0.219	0.406	0.535
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Dep. var.: Full ownership, high income					
Δ MaPP	0.085	-0.038	0.075	0.023	0.344
	(0.106)	(0.138)	(0.179)	(0.224)	(0.256)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.065	0.057	0.154	0.239	0.355
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No
Dep. var.: Ownership with a mortgage					
Δ MaPP	0.005	0.074	0.012	0.119	-0.132
	(0.080)	(0.103)	(0.136)	(0.171)	(0.209)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.013	0.072	0.222	0.410	0.534
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Dep. var.: Full ownership					
Δ MaPP	0.051	-0.028	0.060	-0.004	0.339
	(0.095)	(0.136)	(0.168)	(0.224)	(0.249)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.057	0.073	0.158	0.232	0.351
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No
Dep. var.: Tenant					
Δ MaPP	-0.054	-0.051	-0.068	-0.118	-0.208
	(0.065)	(0.104)	(0.094)	(0.096)	(0.173)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	0.008	0.165	0.331	0.421	0.577
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No

Notes: The dependent variable is the cumulative change in the homeownership rate of each group from the year before the index change, and the independent variable is the change in the macroprudential index. All specifications use the Eurostat control set. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B5: The effect of macroprudential policies on the homeownership rate for lending standard restrictions

(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5

Notes: The dependent variable is the cumulative change in the homeownership rate of each group from the year before the index change, and the independent variable is the change in the index of lending standard restrictions. All specifications use the Eurostat control set. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B6: Implied effect of macroprudential tightening on the aggregate homeownership rate

	Horizon (years)		
Reduction in FTB flows (α)	1	3	5
Panel A: Permanent exclusion			
10%	0.13	0.39	0.65
20%	0.26	0.78	1.30
30%	0.39	1.17	1.95
Panel B: 50% recovery within horizon			
10%	0.07	0.20	0.33
20%	0.13	0.39	0.65
30%	0.20	0.59	0.98

Notes: Implied percentage-point changes in the aggregate homeownership rate, computed from $|\Delta h_h| \approx h \cdot \alpha \cdot \phi$ with first-time buyer share of households $\phi = 1.3\%$. Panel A assumes excluded first-time buyers never enter ownership. Panel B assumes 50% of excluded first-time buyers enter ownership within the five-year horizon (effective α scaled linearly). Compare to the 0.5 percentage point equivalence bound estimated in Table B3 and to the 2.8–3 percentage point steady-state predictions of Kaplan *et al.* (2020) and Garriga & Hedlund (2020).

Table B7: Summary statistics, estimation sample

	Mean	Std. dev.	Median	Min	Max	N
Tenure shares, percent of households (levels)						
Homeownership rate	75.95	9.94	74.50	51.40	97.60	319
Owners with a mortgage	25.48	16.81	20.80	0.60	64.40	319
Outright owners	50.47	24.51	55.00	2.90	96.90	319
Homeownership rate, low income	58.39	18.75	57.50	24.30	97.10	319
Homeownership rate, high income	79.10	8.84	78.20	56.30	97.80	319
Renters at market rates	13.66	10.72	11.90	0.70	40.00	319
Renters at reduced rates or rent-free	10.39	6.17	10.50	0.00	31.70	319
Homeownership rate, young adult-only households	67.06	14.16	67.05	33.00	95.56	319
Young adult-only households owning with a mortgage	21.05	14.38	17.14	0.68	54.88	319
Annual changes in tenure shares, percentage points						
Δ Homeownership rate	-0.11	1.38	-0.10	-5.00	12.60	291
Δ Owners with a mortgage	0.23	1.67	0.30	-7.50	11.10	291
Δ Outright owners	-0.34	1.90	-0.30	-10.70	11.50	291
Δ Homeownership rate, low income	-0.52	2.84	-0.50	-9.60	6.90	291
Δ Homeownership rate, high income	-0.03	1.46	0.00	-5.80	14.20	291
Δ Renters at market rates	0.19	1.68	0.20	-15.60	14.30	291
Δ Homeownership rate, young households	-0.24	2.06	-0.32	-5.72	16.00	291
Δ Young households owning with a mortgage	0.09	1.87	0.10	-5.44	9.31	291
Macroprudential policy index						
Macroprudential index, level	6.20	5.78	4.75	-0.50	22.40	319
Δ MaPP (overall index)	0.77	1.45	0.00	-1.00	10.30	319
Δ LSR (lending standard restrictions)	0.17	0.77	0.00	-1.00	9.00	319
Control variables						
GDP growth per capita (%)	2.44	4.81	2.98	-17.82	31.96	319
Inflation (%)	1.91	2.01	1.68	-1.72	14.19	319
Real house price growth (%)	0.06	8.57	0.96	-49.90	30.70	302
Growth in private debt to GDP (%)	1.14	6.52	0.99	-22.12	27.09	319
Δ 3-month money market rate (pp)	-0.31	1.35	-0.23	-11.10	5.09	316
Observations	319					

Notes: Country-years with non-missing homeownership rate and macroprudential index change, 28 countries, 2004–2017. Tenure shares are in percent of households (or of the relevant group); their annual changes are in percentage points. The macroprudential index is the weighted cumulative count of non-counter-cyclical policy actions described in Section 3; LSR is its lending-standard-restriction component. Control variables are in percent or percentage points.

Table B8: Non-zero changes in the macroprudential index, by country and year

Country	Year	ΔMaPP	LSR	T/L	Country	Year	ΔMaPP	LSR	T/L	Country	Year	ΔMaPP	LSR	T/L
Austria	2008	1.00	1.00	T	Germany	2010	4.10	0.00	T	Poland	2007	4.75	0.00	T
Austria	2010	1.10	0.10	T	Germany	2013	3.00	0.00	T	Poland	2010	3.25	3.25	T
Austria	2011	1.00	1.00	T	Greece	2007	-0.30	0.00	L	Poland	2011	5.05	2.50	T
Austria	2012	3.10	0.10	T	Greece	2009	-1.00	0.00	L	Poland	2013	3.80	2.15	T
Austria	2013	2.25	0.00	T	Greece	2010	0.25	0.00	T	Poland	2014	1.35	0.00	T
Belgium	2006	0.60	0.00	T	Greece	2013	2.25	0.00	T	Poland	2015	0.30	0.00	T
Belgium	2007	0.50	0.00	T	Greece	2014	1.10	0.00	T	Portugal	2007	-0.25	0.00	L
Belgium	2009	1.00	0.00	T	Greece	2015	0.25	0.00	T	Portugal	2008	1.00	0.00	T
Belgium	2010	0.25	0.00	T	Greece	2016	-0.55	0.00	L	Portugal	2009	-0.90	-1.00	L
Belgium	2011	1.25	0.00	T	Hungary	2007	0.45	0.00	T	Portugal	2010	2.00	0.00	T
Belgium	2013	4.75	0.00	T	Hungary	2008	1.00	0.00	T	Portugal	2011	3.50	0.00	T
Bulgaria	2005	1.00	1.00	T	Hungary	2009	5.75	6.00	T	Portugal	2012	0.10	0.00	T
Bulgaria	2008	1.10	0.00	T	Hungary	2010	2.25	1.00	T	Portugal	2013	3.00	0.00	T
Bulgaria	2009	2.00	0.00	T	Hungary	2011	4.75	2.75	T	Portugal	2014	4.25	0.00	T
Bulgaria	2010	0.10	0.00	T	Hungary	2012	1.10	0.10	T	Romania	2008	-0.10	0.00	L
Bulgaria	2013	0.75	0.00	T	Hungary	2013	0.75	0.00	T	Romania	2013	1.75	0.00	T
Bulgaria	2014	1.00	0.00	T	Hungary	2014	1.85	1.35	T	Slovakia	2006	-1.00	0.00	L
Bulgaria	2016	1.00	0.00	T	Hungary	2015	4.00	0.00	T	Slovakia	2007	-0.15	0.00	L
Croatia	2010	3.00	0.00	T	Ireland	2005	1.00	0.00	T	Slovakia	2008	1.75	1.00	T
Croatia	2011	0.25	0.00	T	Ireland	2006	2.50	0.00	T	Slovakia	2010	1.00	0.00	T
Croatia	2012	0.25	0.00	T	Ireland	2009	4.00	0.00	T	Slovakia	2011	0.50	0.00	T
Croatia	2013	3.70	2.00	T	Ireland	2010	3.00	0.00	T	Slovakia	2012	2.00	0.00	T
Croatia	2014	3.00	0.00	T	Ireland	2011	1.10	0.00	T	Slovakia	2013	2.35	0.00	T
Croatia	2015	0.25	0.00	T	Ireland	2013	4.85	0.00	T	Slovakia	2014	10.30	9.00	T
Cyprus	2008	5.00	0.00	T	Italy	2006	0.75	0.00	T	Slovenia	2006	1.65	1.00	T
Cyprus	2009	-0.50	0.00	L	Italy	2013	1.10	0.00	T	Slovenia	2008	1.00	0.00	T
Cyprus	2011	1.00	0.00	T	Italy	2015	1.75	0.00	T	Slovenia	2009	0.10	0.00	T
Cyprus	2013	8.20	0.50	T	Latvia	2007	4.00	4.00	T	Slovenia	2010	1.75	0.00	T
Cyprus	2014	-0.85	0.00	L	Latvia	2009	2.00	2.00	T	Slovenia	2011	-0.45	0.00	L
Cyprus	2015	-0.75	0.00	L	Latvia	2013	2.00	0.00	T	Slovenia	2012	1.65	0.00	T
Czechia	2007	0.10	0.00	T	Latvia	2014	0.75	-0.25	T	Slovenia	2013	2.10	0.00	T
Czechia	2013	1.75	0.00	T	Lithuania	2006	-0.15	0.00	L	Slovenia	2014	-0.10	0.00	L
Czechia	2014	4.25	0.00	T	Lithuania	2008	1.00	0.00	T	Spain	2007	0.20	0.00	T
Czechia	2015	2.00	2.00	T	Lithuania	2010	1.00	0.00	T	Spain	2011	2.00	0.00	T
Czechia	2016	0.75	0.50	T	Lithuania	2011	3.00	3.00	T	Sweden	2004	2.00	0.00	T
Denmark	2006	-0.50	0.00	L	Lithuania	2013	2.50	0.00	T	Sweden	2009	1.25	0.00	T
Denmark	2010	5.00	0.00	T	Lithuania	2014	0.90	0.00	T	Sweden	2010	1.00	1.00	T
Denmark	2013	2.10	0.00	T	Lithuania	2015	1.00	0.00	T	Sweden	2011	0.25	0.00	T
Denmark	2014	0.50	0.00	T	Luxembourg	2007	1.00	0.00	T	Sweden	2012	3.00	0.00	T
Estonia	2007	-0.50	0.00	L	Luxembourg	2009	1.10	0.00	T	Sweden	2013	3.00	0.00	T
Estonia	2010	-0.25	0.00	L	Luxembourg	2011	1.00	0.00	T	Sweden	2014	3.25	0.00	T
Estonia	2013	2.65	0.00	T	Luxembourg	2012	2.25	1.00	T	Sweden	2015	2.00	2.00	T
Estonia	2014	1.00	0.00	T	Luxembourg	2013	2.00	0.00	T	Sweden	2016	0.25	0.00	T
Finland	2006	0.95	0.00	T	Luxembourg	2015	3.00	0.00	T	Sweden	2017	1.00	1.00	T
Finland	2010	3.00	1.00	T	Malta	2008	0.50	0.00	T	UK	2006	0.10	0.00	T
Finland	2011	-0.50	0.00	L	Malta	2012	1.00	0.00	T	UK	2009	0.90	0.00	T
Finland	2012	1.00	0.00	T	Malta	2013	6.10	0.00	T	UK	2010	1.25	0.00	T
Finland	2013	1.25	0.00	T	Netherlands	2009	0.10	0.00	T	UK	2011	2.25	0.00	T
Finland	2014	1.00	0.00	T	Netherlands	2010	0.25	0.00	T	UK	2012	0.25	0.00	T
Finland	2015	1.00	0.00	T	Netherlands	2011	3.00	2.00	T	UK	2013	4.20	0.00	T
France	2012	1.00	0.00	T	Netherlands	2012	2.75	2.75	T	UK	2014	2.10	2.00	T
France	2013	3.50	0.00	T	Netherlands	2013	2.00	0.00	T	UK	2015	1.35	0.00	T
France	2016	0.10	0.00	T	Netherlands	2014	4.25	0.00	T					

Notes: All country-years in the estimation sample with a non-zero change in the macroprudential index, in three column groups. ΔMaPP is the change in the overall index; LSR is the component from loan-to-value, debt-service-to-income and related borrower-based limits, so the difference is the contribution of capital, liquidity, leverage and other instruments. T denotes a net tightening and L a net loosening. Weights: 1 for activation of a tool, 0.25 for a change in level, 0.1 for a change in scope, 0.05 for maintenance (Table B1). Actions coded as counter-cyclical in MaPPED are excluded, which is why the Irish 2015 and Finnish 2016 reforms do not appear.

Table B9: Estimation sample under alternative sets of control variables

	h=1	h=2	h=3	h=4	h=5
No controls					
Observations	319	291	263	235	207
Countries	28	28	28	28	28
BIS credit and policy rate					
Observations	181	163	145	127	109
Countries	18	18	18	18	18
Eurostat private debt and money market rate					
Observations	272	245	218	190	162
Countries	28	28	28	28	28

Notes: Number of country-year observations and countries in the local projection for the aggregate homeownership rate at each horizon. The BIS control set uses the change in BIS credit to the private non-financial sector relative to GDP and the change in the BIS central bank policy rate. The Eurostat control set uses the change in Eurostat private sector debt relative to GDP and the change in the 3-month money market rate. Both sets also include lagged GDP growth per capita, inflation, real house price growth and the level of the macroprudential index. See Appendix C for sources.

Table B10: The effect of macroprudential policies on the homeownership rate: alternative control sets

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Panel A: No controls					
Δ MaPP	0.041	0.006	-0.043	-0.034	-0.083
	(0.079)	(0.098)	(0.100)	(0.107)	(0.156)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
TOST p-value (δ = .5 pp)	0.000	0.000	0.000	0.000	0.006
Equivalence at δ = .5 pp	Yes	Yes	Yes	Yes	Yes
Minimum detectable effect (pp)	0.222	0.274	0.280	0.300	0.438
Effect of a one s.d. change (1.45 units, pp)	0.059	0.008	-0.062	-0.050	-0.120
Effect of the median tightening (1.30 units, pp)	0.053	0.008	-0.056	-0.045	-0.107
Panel B: Eurostat controls (all 28 countries)					
Δ MaPP	0.054	0.051	0.068	0.118	0.208
	(0.065)	(0.104)	(0.094)	(0.096)	(0.173)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
TOST p-value (δ = .5 pp)	0.000	0.000	0.000	0.000	0.051
Equivalence at δ = .5 pp	Yes	Yes	Yes	Yes	No
Minimum detectable effect (pp)	0.183	0.292	0.264	0.268	0.485
Effect of a one s.d. change (1.45 units, pp)	0.078	0.073	0.098	0.170	0.300
Effect of the median tightening (1.30 units, pp)	0.071	0.066	0.088	0.153	0.270
Panel C: BIS controls					
Δ MaPP	0.124	0.248	0.237	0.315**	0.526***
	(0.109)	(0.155)	(0.143)	(0.130)	(0.141)
Observations	181	163	145	127	109
Countries	18	18	18	18	18
TOST p-value (δ = .5 pp)	0.002	0.061	0.042	0.086	0.571
Equivalence at δ = .5 pp	Yes	No	Yes	No	No
Minimum detectable effect (pp)	0.307	0.435	0.402	0.364	0.395
Effect of a one s.d. change (1.45 units, pp)	0.179	0.358	0.343	0.456	0.761
Effect of the median tightening (1.30 units, pp)	0.161	0.322	0.308	0.410	0.683

Notes: Local projections of the cumulative change in the aggregate homeownership rate, from the year before the index change to horizon h, on the change in the macroprudential policy index, Equation (1). The last two rows multiply the coefficient by the standard deviation of the index change and by the median non-zero tightening in the estimation sample. Panel A includes country and year fixed effects only. Panel B adds lagged GDP growth per capita, inflation, real house price growth, growth in Eurostat private sector debt to GDP, the change in the 3-month money market rate and the level of the index; these controls are available for all 28 countries. Panel C uses the BIS credit and policy rate series instead, which are missing for ten countries. The TOST p-value is the larger of the two one-sided p-values against $\pm\delta$; equivalence holds when it is below 0.05. The minimum detectable effect is the true effect that would be detected with 80 percent power at the 5 percent level. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B11: Equivalence tests for alternative margins and minimum detectable effects

	h=1	h=2	h=3	h=4	h=5
Panel A: No controls					
Point estimate	0.041	0.006	-0.043	-0.034	-0.083
Standard error	0.079	0.098	0.100	0.107	0.156
Minimum detectable effect (80% power)	0.222	0.274	0.280	0.300	0.438
TOST p-values by equivalence margin δ (implied permanent FTB exclusion α at $h = 5$)					
$\delta = 0.10$ pp ($\alpha = 1.5\%$)	0.232	0.172	0.287	0.273	0.456
$\delta = 0.25$ pp ($\alpha = 3.8\%$)	¡0.01	¡0.01	0.024	0.027	0.147
$\delta = 0.50$ pp ($\alpha = 7.7\%$)	¡0.01	¡0.01	¡0.01	¡0.01	¡0.01
$\delta = 1.00$ pp ($\alpha = 15.4\%$)	¡0.01	¡0.01	¡0.01	¡0.01	¡0.01
Panel B: Eurostat controls (all 28 countries)					
Point estimate	0.054	0.051	0.068	0.118	0.208
Standard error	0.065	0.104	0.094	0.096	0.173
Minimum detectable effect (80% power)	0.183	0.292	0.264	0.268	0.485
TOST p-values by equivalence margin δ (implied permanent FTB exclusion α at $h = 5$)					
$\delta = 0.10$ pp ($\alpha = 1.5\%$)	0.245	0.320	0.368	0.573	0.731
$\delta = 0.25$ pp ($\alpha = 3.8\%$)	¡0.01	0.033	0.032	0.089	0.404
$\delta = 0.50$ pp ($\alpha = 7.7\%$)	¡0.01	¡0.01	¡0.01	¡0.01	0.051
$\delta = 1.00$ pp ($\alpha = 15.4\%$)	¡0.01	¡0.01	¡0.01	¡0.01	¡0.01

Notes: Two one-sided tests (TOST) of the null hypothesis that the effect of a one-unit change in the macroprudential index on the aggregate homeownership rate lies outside $[-\delta, \delta]$, for the estimates in Table B10. Equivalence at the 5 percent level is established when the reported p-value is below 0.05. The implied α is the permanent reduction in the flow of first-time buyers that would produce an aggregate effect of δ after five years under the stock-flow benchmark of Section 10, $|\Delta h| = h \cdot \alpha \cdot \phi$ with $\phi = 1.3$ percent. The minimum detectable effect is 2.8 times the standard error (80 percent power, 5 percent two-sided test).

Table B12: First stage: the response of house prices and credit to the macroprudential index

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Real house prices (log x 100), no controls					
Δ MaPP	-0.137	-0.515	-1.020*	-0.953	-1.405
	(0.248)	(0.407)	(0.555)	(0.701)	(0.843)
Observations	302	274	246	218	190
Countries	28	28	28	28	28
Real house prices (log x 100), with controls					
Δ MaPP	-0.186	-0.483	-0.418	0.147	0.577
	(0.209)	(0.411)	(0.593)	(0.703)	(0.929)
Observations	279	252	225	197	169
Countries	28	28	28	28	28
Private debt to GDP (log x 100, Eurostat), no controls					
Δ MaPP	0.285	0.093	0.497	0.425	0.151
	(0.229)	(0.288)	(0.341)	(0.360)	(0.490)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Private debt to GDP (log x 100, Eurostat), with controls					
Δ MaPP	0.150	0.008	0.432	0.355	0.035
	(0.248)	(0.292)	(0.322)	(0.257)	(0.410)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Household debt to GDP (log x 100, Eurostat), no controls					
Δ MaPP	0.357	0.416	0.382	0.110	-0.514
	(0.240)	(0.334)	(0.352)	(0.453)	(0.675)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Household debt to GDP (log x 100, Eurostat), with controls					
Δ MaPP	0.240	0.338	0.241	-0.283	-1.591*
	(0.230)	(0.330)	(0.440)	(0.570)	(0.919)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Real household debt (log x 100, Eurostat), no controls					
Δ MaPP	0.054	-0.064	-0.163	-0.470	-0.585
	(0.221)	(0.266)	(0.262)	(0.348)	(0.657)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Real household debt (log x 100, Eurostat), with controls					
Δ MaPP	-0.051	-0.007	0.064	-0.383	-1.006
	(0.214)	(0.285)	(0.360)	(0.455)	(0.722)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Credit to private non-financial sector to GDP (log x 100, BIS), no controls					
Δ MaPP	0.247	-0.072	0.055	-0.116	-0.413
	(0.243)	(0.436)	(0.696)	(0.580)	(0.359)
Observations	211	193	175	157	139
Countries	18	18	18	18	18
Credit to private non-financial sector to GDP (log x 100, BIS), with controls					
Δ MaPP	0.006	-0.519	-0.386	-0.853	-1.458*
	(0.274)	(0.447)	(0.526)	(0.572)	(0.779)
Observations	181	163	145	127	109
Countries	18	18	18	18	18

Notes: Cumulative local projections, $y_{i,t+h} - y_{i,t-1}$, of the log of each variable (times 100) on the change in the macroprudential index, with country and year fixed effects. Coefficients are percent changes. Specifications with controls include lagged GDP growth per capita, inflation, the change in the 3-month money market rate, the level of the index, and the lagged growth of the other first-stage variable (private debt for house prices; house prices for the debt series). Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B13: First stage and homeownership response for lending standard restrictions

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Real house prices (log x 100), no controls					
Δ LSR	-0.539	-0.868	-0.903	-0.858	-1.773
	(0.557)	(0.765)	(0.906)	(1.001)	(1.563)
Observations	302	274	246	218	190
Countries	28	28	28	28	28
Real house prices (log x 100), with controls					
Δ LSR	-0.377	-0.600	-0.250	-0.012	0.628
	(0.294)	(0.600)	(0.690)	(0.682)	(1.436)
Observations	279	252	225	197	169
Countries	28	28	28	28	28
Private debt to GDP (log x 100), no controls					
Δ LSR	0.529	0.623	1.116**	0.732	0.217
	(0.363)	(0.493)	(0.485)	(0.649)	(1.067)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Private debt to GDP (log x 100), with controls					
Δ LSR	0.417	0.651	1.339**	1.168**	1.488
	(0.317)	(0.512)	(0.551)	(0.429)	(0.932)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Household debt to GDP (log x 100), no controls					
Δ LSR	0.355	0.325	0.331	-0.304	-2.180**
	(0.326)	(0.460)	(0.544)	(0.880)	(1.033)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Household debt to GDP (log x 100), with controls					
Δ LSR	0.311	0.472	0.852*	0.426	-1.702
	(0.202)	(0.381)	(0.476)	(0.638)	(1.934)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Real household debt (log x 100), no controls					
Δ LSR	0.022	-0.049	-0.241	-1.027	-1.995
	(0.397)	(0.473)	(0.386)	(0.720)	(1.381)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Real household debt (log x 100), with controls					
Δ LSR	0.103	0.456	0.699*	-0.068	-1.425
	(0.241)	(0.372)	(0.351)	(0.514)	(1.188)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Homeownership rate (pp), no controls					
Δ LSR	0.173	-0.001	-0.069	-0.130	-0.382
	(0.207)	(0.141)	(0.093)	(0.120)	(0.313)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Homeownership rate (pp), with controls					
Δ LSR	0.039	-0.006	0.058	0.102	0.360
	(0.080)	(0.121)	(0.137)	(0.191)	(0.353)
Observations	272	245	218	190	162
Countries	28	28	28	28	28

Notes: As Table B12, but the shock is the change in the index of lending standard restrictions (loan-to-value, debt-service-to-income and related borrower-based limits) rather than the overall index. The final rows report the response of the aggregate homeownership rate in percentage points. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B14: Pre-trends: past changes in the homeownership rate and the policy index

	(1) Δ Owner$_{t-3}$	(2) Δ Owner$_{t-2}$	(3) Δ Owner$_{t-1}$
Panel A: No controls			
Δ MaPP$_t$	0.035	-0.019	-0.002
	(0.063)	(0.049)	(0.120)
Observations	235	263	291
Countries	28	28	28
Joint test: lagged changes in ownership (t-1 to t-3) predict Δ MaPP$_t$, p-value		0.674	
Panel B: Eurostat controls			
Δ MaPP$_t$	-0.004	-0.127*	-0.119*
	(0.079)	(0.072)	(0.067)
Observations	229	251	272
Countries	28	28	28
Joint test: lagged changes in ownership (t-1 to t-3) predict Δ MaPP$_t$, p-value		0.222	

Notes: Each column regresses the change in the homeownership rate in the indicated year before the policy change on the change in the macroprudential index in year t, with country and year fixed effects. The joint test is the p-value of a Wald test that the three lagged changes in the homeownership rate have zero coefficients in a regression of the index change on those lags. Panel B adds the Eurostat control set. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B15: Randomization inference for the aggregate local projection

	h=1	h=2	h=3	h=4	h=5
Observed coefficient	0.041	0.006	-0.043	-0.034	-0.083
Randomization-inference p-value	0.512	0.954	0.678	0.781	0.568
97.5th percentile of permuted coefficients	0.132	0.171	0.193	0.225	0.279
Joint p-value, largest absolute coefficient across horizons			0.807		
Permutations			2000		

Notes: The macroprudential index change is reshuffled across countries within each year and the cumulative local projection without controls is re-estimated, 2,000 times. The p-value is the share of permutations with an absolute coefficient at least as large as the observed one. The joint p-value uses the largest absolute coefficient across the five horizons.

Table B16: The effect of macroprudential policies on the homeownership rate of young households

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Panel A: No controls					
Δ MaPP	0.118	0.039	-0.163	-0.149	-0.094
	(0.095)	(0.149)	(0.143)	(0.164)	(0.228)
Observations	319	291	263	235	207
Economic significance	0.08	0.02	0.06	0.05	0.03
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Panel B: With controls					
Δ MaPP	0.104	0.028	-0.080	0.046	0.384**
	(0.093)	(0.136)	(0.174)	(0.177)	(0.172)
L.Δ GDP per capita	12.169***	4.187	10.073	15.252	15.875
	(4.023)	(5.573)	(10.335)	(12.711)	(10.303)
L.Inflation	6.888	16.517	15.074	-3.958	-28.940**
	(7.266)	(11.956)	(11.446)	(17.015)	(14.000)
L.Δ House prices	-1.965	2.496	5.419*	5.079	2.709
	(1.825)	(2.643)	(3.115)	(4.311)	(4.196)
L.Δ Private debt to GDP	4.978**	1.600	5.722*	6.545*	11.394*
	(1.923)	(2.524)	(3.046)	(3.285)	(5.632)
L.Δ 3-month money market rate	0.090	0.276***	0.254	0.173	0.343***
	(0.103)	(0.083)	(0.151)	(0.105)	(0.105)
L.Macroprudential policy index (MaPP)	-0.005	-0.054	0.038	0.214	0.360
	(0.043)	(0.107)	(0.159)	(0.189)	(0.232)
Observations	272	245	218	190	162
Economic significance	0.07	0.01	0.03	0.02	0.12
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No

Notes: As Table B3, with the cumulative change in the homeownership rate of adult-only households under 65 as the dependent variable. Panel B uses the Eurostat control set. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B17: The effect of macroprudential policies on homeownership rates by age group

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Dep. var.: Homeownership rate, adult-only HH under 65					
Δ MaPP	0.104	0.028	-0.080	0.046	0.384**
	(0.093)	(0.136)	(0.174)	(0.177)	(0.172)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.015	0.082	0.216	0.362	0.603
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No
Dep. var.: Homeownership rate, adult-only HH 65+					
Δ MaPP	-0.020	0.046	0.038	0.058	0.260
	(0.072)	(0.164)	(0.137)	(0.142)	(0.156)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.044	0.098	0.237	0.383	0.515
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	No
Dep. var.: Homeownership rate, age 18-54					
Δ MaPP	0.092	0.040	0.048	0.088	0.126
	(0.072)	(0.108)	(0.102)	(0.104)	(0.170)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	0.023	0.212	0.403	0.516	0.633
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Dep. var.: Homeownership rate, age 25-54					
Δ MaPP	0.082	0.031	0.047	0.096	0.102
	(0.068)	(0.105)	(0.106)	(0.097)	(0.152)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	0.040	0.251	0.405	0.527	0.666
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Dep. var.: Homeownership rate, age 55-64					
Δ MaPP	0.097	0.045	0.059	-0.006	0.044
	(0.093)	(0.100)	(0.124)	(0.113)	(0.155)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.034	0.079	0.179	0.333	0.484
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes
Dep. var.: Homeownership rate, age 65+					
Δ MaPP	-0.111	-0.032	-0.006	0.025	0.145
	(0.065)	(0.164)	(0.111)	(0.132)	(0.157)
Control variables	Yes	Yes	Yes	Yes	Yes
Observations	272	245	218	190	162
Adjusted R^2	-0.016	0.141	0.305	0.426	0.536
Equivalence (CI fully inside margin):	Yes	Yes	Yes	Yes	Yes

Notes: Cumulative local projections with the Eurostat control set for the homeownership rate of adult-only households under 65 and 65 and over (EU-SILC tenure by household composition), and of individuals aged 18–54, 25–54, 55–64 and 65 and over (EU-SILC tenure by age). Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B18: The effect of macroprudential policies on the share of young households owning with a mortgage

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Panel A: No controls					
Δ MaPP	0.161**	0.043	-0.037	0.113	-0.119
	(0.078)	(0.087)	(0.130)	(0.148)	(0.171)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Equivalence at $\delta = 0.5$ pp	Yes	Yes	Yes	Yes	Yes
Equivalence at $\delta = 1.0$ pp	Yes	Yes	Yes	Yes	Yes
Minimum detectable effect (pp)	0.219	0.244	0.363	0.413	0.480
Panel B: Eurostat controls (all 28 countries)					
Δ MaPP	0.121	0.047	0.064	0.282	0.006
	(0.111)	(0.139)	(0.170)	(0.177)	(0.228)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = 0.5$ pp	Yes	Yes	Yes	No	Yes
Equivalence at $\delta = 1.0$ pp	Yes	Yes	Yes	Yes	Yes
Minimum detectable effect (pp)	0.310	0.388	0.477	0.496	0.638

Notes: The dependent variable is the share of adult-only households under 65 who own their home with an outstanding mortgage, from EU-SILC. The share is a stock; among the available cross-country series it is the one most exposed to first-time-buyer entry. Cumulative responses. Panel B uses the Eurostat control set. Equivalence is established at the 5 percent level when the 90 percent confidence interval lies within $\pm\delta$. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B19: The effect of macroprudential policies on the rental market

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Dep. var.: Renters					
Δ MaPP	-0.054	-0.051	-0.068	-0.118	-0.208
	(0.065)	(0.104)	(0.094)	(0.096)	(0.173)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp	Yes	Yes	Yes	Yes	No
Dep. var.: Renters, market rate					
Δ MaPP	-0.122	-0.386	-0.349	-0.203	-0.348*
	(0.079)	(0.281)	(0.215)	(0.122)	(0.183)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp	Yes	No	No	Yes	No
Dep. var.: Renters, reduced rate or free					
Δ MaPP	0.064	0.333	0.280	0.084	0.148
	(0.089)	(0.227)	(0.202)	(0.101)	(0.129)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp	Yes	No	No	Yes	Yes
Dep. var.: Renter, adult-only HH under 65					
Δ MaPP	-0.138	-0.006	0.035	-0.078	-0.349*
	(0.105)	(0.141)	(0.172)	(0.175)	(0.186)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp	Yes	Yes	Yes	Yes	No
Dep. var.: Market-rate renter, adult-only HH under 65					
Δ MaPP	-0.169	-0.361	-0.323	-0.151	-0.433*
	(0.136)	(0.342)	(0.292)	(0.191)	(0.251)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp	Yes	No	No	Yes	No
Dep. var.: Reduced-rent/free renter, adult-only HH under 65					
Δ MaPP	0.060	0.324	0.431*	0.092	0.102
	(0.099)	(0.286)	(0.234)	(0.111)	(0.165)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp	Yes	No	No	Yes	Yes

Notes: Cumulative local projections with the Eurostat control set. The first three dependent variables are the shares of all households renting, renting at market rates, and renting at reduced rates or rent-free. The last three are the same shares among adult-only households under 65. All variables are in percentage points of the relevant population, from EU-SILC. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B20: Homeownership response to tightenings that were followed by a fall in household credit

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Dep. var.: homeownership rate; effective = credit fell after the tightening (51 of 127 tightenings); no controls					
Δ MaPP, credit fell	0.125	0.129	0.156	0.160	0.076
	(0.078)	(0.118)	(0.128)	(0.098)	(0.146)
Δ MaPP, credit did not fall	-0.002	-0.072	-0.180	-0.172	-0.256
	(0.110)	(0.104)	(0.118)	(0.152)	(0.296)
Observations	314	287	263	235	207
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	Yes	Yes	Yes	Yes
Minimum detectable effect (pp)	0.219	0.330	0.358	0.274	0.408
Dep. var.: homeownership rate; effective = credit fell after the tightening (51 of 127 tightenings); Eurostat controls					
Δ MaPP, credit fell	0.165**	0.219*	0.254**	0.294**	0.273
	(0.076)	(0.114)	(0.115)	(0.132)	(0.217)
Δ MaPP, credit did not fall	-0.036	-0.103	-0.113	-0.050	0.056
	(0.065)	(0.100)	(0.093)	(0.097)	(0.159)
Observations	268	242	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	Yes	Yes	No	No
Minimum detectable effect (pp)	0.214	0.319	0.323	0.369	0.608
Dep. var.: homeownership rate; effective = credit decelerated after the tightening (67 of 114 tightenings); no controls					
Δ MaPP, credit fell	0.195*	0.184	0.140	0.148	0.033
	(0.114)	(0.123)	(0.121)	(0.115)	(0.151)
Δ MaPP, credit did not fall	-0.033	-0.101	-0.122	-0.118	-0.048
	(0.064)	(0.107)	(0.125)	(0.147)	(0.352)
Observations	301	274	250	222	193
Countries	28	28	28	28	27
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	Yes	Yes	Yes	Yes
Minimum detectable effect (pp)	0.319	0.346	0.339	0.323	0.422
Dep. var.: homeownership rate; effective = credit decelerated after the tightening (67 of 114 tightenings); Eurostat controls					
Δ MaPP, credit fell	0.148*	0.223*	0.257**	0.297**	0.251
	(0.083)	(0.124)	(0.115)	(0.124)	(0.207)
Δ MaPP, credit did not fall	-0.011	-0.072	-0.066	-0.009	0.092
	(0.062)	(0.096)	(0.082)	(0.091)	(0.189)
Observations	262	236	212	184	155
Countries	28	28	28	28	27
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	Yes	Yes	No	No
Minimum detectable effect (pp)	0.231	0.346	0.321	0.347	0.580
Dep. var.: young households owning with a mortgage; credit fell; Eurostat controls					
Δ MaPP, credit fell	0.089	0.012	0.091	0.265	0.121
	(0.124)	(0.197)	(0.184)	(0.190)	(0.216)
Δ MaPP, credit did not fall	0.164	0.085	0.028	0.301	-0.298
	(0.135)	(0.139)	(0.223)	(0.254)	(0.444)
Observations	268	242	218	190	162
Countries	28	28	28	28	28
Dep. var.: renters at market rates; credit fell; Eurostat controls					
Δ MaPP, credit fell	-0.150	-0.551	-0.484	-0.180	-0.371*
	(0.102)	(0.417)	(0.324)	(0.139)	(0.213)
Δ MaPP, credit did not fall	-0.096	-0.212	-0.236	-0.227	-0.336
	(0.080)	(0.192)	(0.189)	(0.203)	(0.254)
Observations	268	242	218	190	162
Countries	28	28	28	28	28

Notes: Each tightening of the macroprudential index (index change greater than zero) is classified by the cumulative change in real household debt between $t-1$ and $t+2$. "Credit fell" means the change was negative; "credit decelerated" means it was smaller than the change over the three preceding years. The two components enter the same local projection, so each coefficient compares country-years with that type of tightening to country-years without a tightening. Tightenings in the last two sample years cannot be classified and are excluded. Classification uses a post-treatment outcome, so the comparison is descriptive. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B21: First stage for tightenings followed by a fall in household credit

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Real household debt (log x 100), cumulative					
Δ MaPP, credit fell	-0.434	-1.072**	-1.569***	-2.091***	-2.012**
	(0.335)	(0.402)	(0.456)	(0.643)	(0.915)
Δ MaPP, credit did not fall	0.519*	0.858	1.070	0.926	1.855**
	(0.264)	(0.547)	(0.677)	(0.751)	(0.713)
Observations	314	287	263	235	207
Countries	28	28	28	28	28
Real house prices (log x 100), cumulative					
Δ MaPP, credit fell	-0.652*	-1.574**	-2.195***	-1.931**	-1.194
	(0.382)	(0.616)	(0.784)	(0.806)	(0.911)
Δ MaPP, credit did not fall	0.430	0.364	-0.144	-0.267	-1.921
	(0.319)	(0.462)	(0.507)	(0.744)	(1.355)
Observations	297	270	246	218	190
Countries	28	28	28	28	28

Notes: Cumulative responses (percent) of real household debt and real house prices to the two components of the macroprudential index defined in Table B20, without controls. The household debt response to "credit fell" tightenings at horizons one to three is negative by construction. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B22: Homeownership response to lending standard restrictions that were followed by a fall in household credit

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Effective = credit fell (14 of 28 tightenings); no controls					
Δ LSR, credit fell	0.029	-0.040	0.006	-0.127	-0.150
	(0.170)	(0.154)	(0.150)	(0.228)	(0.237)
Δ LSR, credit did not fall	0.265	0.027	-0.100	-0.126	-1.197**
	(0.349)	(0.235)	(0.097)	(0.129)	(0.497)
Observations	317	290	263	235	207
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	Yes	Yes	No	No
Memo: real household debt response to effective tightenings (%)	-1.163	-1.564	-1.765	-3.263	-3.483
Effective = credit fell (14 of 28 tightenings); Eurostat controls					
Δ LSR, credit fell	0.034	0.106	0.215	0.188	0.299
	(0.182)	(0.224)	(0.267)	(0.401)	(0.392)
Δ LSR, credit did not fall	0.046	-0.066	-0.020	0.063	0.744**
	(0.079)	(0.089)	(0.082)	(0.169)	(0.344)
Observations	270	244	218	190	162
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	Yes	No	No	No
Memo: real household debt response to effective tightenings (%)	-1.006	-1.089	-0.787	-2.184	-2.618
Effective = credit decelerated (17 of 24 tightenings); no controls					
Δ LSR, credit fell	0.576	0.461	0.293*	0.307***	0.027
	(0.454)	(0.304)	(0.167)	(0.105)	(0.150)
Δ LSR, credit did not fall	-0.058	-0.192***	-0.198**	-0.239	-1.319*
	(0.057)	(0.062)	(0.083)	(0.172)	(0.766)
Observations	313	286	259	231	203
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp, effective tightenings	No	No	No	Yes	Yes
Memo: real household debt response to effective tightenings (%)	0.256	0.242	-0.498	-1.964	-2.689
Effective = credit decelerated (17 of 24 tightenings); Eurostat controls					
Δ LSR, credit fell	0.106	0.262	0.356	0.575	0.610*
	(0.180)	(0.205)	(0.254)	(0.343)	(0.321)
Δ LSR, credit did not fall	0.009	-0.086	-0.019	-0.022	-0.064
	(0.062)	(0.064)	(0.070)	(0.113)	(0.475)
Observations	269	243	217	189	161
Countries	28	28	28	28	28
Equivalence at $\delta = .5$ pp, effective tightenings	Yes	No	No	No	No
Memo: real household debt response to effective tightenings (%)	-0.627	-0.648	-0.470	-2.170	-2.885

Notes: As Table B20, using the index of lending standard restrictions in place of the overall index. The memo row reports the cumulative percent response of real household debt to the effective tightenings from the same specification. Standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B23: Homeownership response by pre-policy depth of the mortgage market

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Homeownership rate (pp), no controls					
Δ MaPP, household debt above median	0.074	0.171	0.188	0.237	0.147
	(0.089)	(0.143)	(0.151)	(0.150)	(0.193)
Δ MaPP, household debt below median	0.001	-0.195**	-0.330**	-0.372*	-0.592
	(0.124)	(0.093)	(0.130)	(0.186)	(0.369)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Homeownership rate (pp), Eurostat controls					
Δ MaPP, household debt above median	0.116	0.153	0.140	0.164*	0.166
	(0.083)	(0.131)	(0.117)	(0.092)	(0.160)
Δ MaPP, household debt below median	-0.040	-0.105	-0.039	0.051	0.414
	(0.077)	(0.109)	(0.106)	(0.155)	(0.331)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Homeownership rate (pp), continuous interaction, Eurostat controls					
Δ MaPP x standardized household debt to GDP	0.033	0.075	0.066	0.071	-0.036
	(0.033)	(0.057)	(0.039)	(0.044)	(0.078)
Δ MaPP	0.058	0.051	0.100	0.165	0.338
	(0.070)	(0.100)	(0.100)	(0.107)	(0.219)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Real household debt (log x 100), no controls					
Δ MaPP, household debt above median	0.607*	1.072**	1.052*	0.651	0.443
	(0.323)	(0.507)	(0.580)	(0.545)	(0.651)
Δ MaPP, household debt below median	-0.612	-1.450**	-1.670***	-1.863***	-2.869**
	(0.450)	(0.567)	(0.573)	(0.663)	(1.325)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Real house prices (log x 100), no controls					
Δ MaPP, household debt above median	0.037	-0.413	-1.282	-1.424	-1.755
	(0.304)	(0.600)	(0.808)	(0.925)	(1.103)
Δ MaPP, household debt below median	-0.368	-0.654	-0.655	-0.292	-0.408
	(0.328)	(0.485)	(0.681)	(0.923)	(1.541)
Observations	302	274	246	218	190
Countries	28	28	28	28	28

Notes: The change in the macroprudential index is split according to whether household debt to GDP in the year before the change was above or below the sample median; the two components enter the same regression. Median household debt to GDP in the year before the policy change: 51.9 percent. The continuous specification interacts the index change with standardized lagged household debt to GDP and includes both main effects. Household debt to GDP is measured before the policy change, so the split does not condition on outcomes. Cumulative responses; standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B24: Tightenings and loosenings separately

	(1) h=1	(2) h=2	(3) h=3	(4) h=4	(5) h=5
Overall index, no controls (132 tightening and 17 loosening country-years)					
Δ MaPP, tightenings	0.076	0.038	-0.015	-0.011	-0.028
	(0.076)	(0.095)	(0.101)	(0.102)	(0.160)
Δ MaPP, loosenings	-1.329	-1.239	-1.076	-0.893	-2.128
	(0.955)	(0.941)	(1.100)	(1.270)	(1.297)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Overall index, Eurostat controls					
Δ MaPP, tightenings	0.060	0.051	0.059	0.108	0.214
	(0.069)	(0.107)	(0.095)	(0.094)	(0.168)
Δ MaPP, loosenings	-0.340	0.032	0.710	0.968	-0.389
	(0.460)	(0.426)	(0.648)	(1.171)	(1.273)
Observations	272	245	218	190	162
Countries	28	28	28	28	28
Lending standard restrictions, no controls (30 tightening and 2 loosening country-years)					
Δ LSR, tightenings	0.178	0.002	-0.059	-0.126	-0.392
	(0.210)	(0.142)	(0.095)	(0.120)	(0.322)
Δ LSR, loosenings	-0.427	-0.358	-1.373	-0.690	0.302
	(0.284)	(0.545)	(0.924)	(1.293)	(0.528)
Observations	319	291	263	235	207
Countries	28	28	28	28	28
Lending standard restrictions, Eurostat controls					
Δ LSR, tightenings	0.042	-0.005	0.066	0.116	0.360
	(0.080)	(0.121)	(0.136)	(0.192)	(0.353)
Δ LSR, loosenings	-2.766*	-1.620	-7.644***	-11.122***	0.000
	(1.485)	(2.431)	(2.697)	(3.810)	(.)
Observations	272	245	218	190	162
Countries	28	28	28	28	28

Notes: The index change is split into its positive part (tightenings) and negative part (loosenings); both enter the same regression, so each coefficient is measured against country-years with no change. A loosening coefficient of β means that a one-unit loosening changes the homeownership rate by $-\beta$. Cumulative responses of the aggregate homeownership rate; standard errors clustered by country in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B25: Alternative inference for the aggregate cumulative response

	h=1	h=2	h=3	h=4	h=5
Panel A: No controls					
Coefficient (pp)	0.041	0.006	-0.043	-0.034	-0.083
Clustered s.e.	(0.079)	(0.098)	(0.100)	(0.107)	(0.156)
Clustered p-value	0.610	0.953	0.669	0.750	0.602
Wild cluster bootstrap p-value	0.621	0.955	0.695	0.756	0.641
Wild cluster bootstrap 95% CI	[-0.110, 0.240]	[-0.192, 0.242]	[-0.251, 0.196]	[-0.257, 0.214]	[-0.416, 0.252]
Driscoll-Kraay s.e.	(0.074)	(0.085)	(0.133)	(0.152)	(0.195)
Leave-one-country-out range	[-0.006, 0.082]	[-0.037, 0.063]	[-0.092, -0.000]	[-0.072, 0.018]	[-0.126, 0.031]
Population-weighted coefficient	0.110	-0.057	-0.138	-0.126	-0.239
Population-weighted s.e.	(0.172)	(0.099)	(0.098)	(0.160)	(0.277)
Panel B: Eurostat controls					
Coefficient (pp)	0.054	0.051	0.068	0.118	0.208
Clustered s.e.	(0.065)	(0.104)	(0.094)	(0.096)	(0.173)
Clustered p-value	0.414	0.632	0.477	0.229	0.240
Wild cluster bootstrap p-value	0.422	0.683	0.495	0.227	0.372
Wild cluster bootstrap 95% CI	[-0.070, 0.211]	[-0.143, 0.307]	[-0.113, 0.292]	[-0.065, 0.373]	[-0.183, 0.664]
Driscoll-Kraay s.e.	(0.061)	(0.073)	(0.073)	(0.061)	(0.077)
Leave-one-country-out range	[0.013, 0.081]	[-0.005, 0.116]	[0.009, 0.122]	[0.068, 0.190]	[0.001, 0.295]
Population-weighted coefficient	-0.003	0.008	0.066	0.151	0.262
Population-weighted s.e.	(0.081)	(0.110)	(0.098)	(0.104)	(0.132)

Notes: Cumulative response of the aggregate homeownership rate to a one-unit change in the macroprudential index. The wild cluster bootstrap uses 999 replications with Webb weights, clustered by country (Roodman *et al.* , 2019). Driscoll-Kraay standard errors allow for cross-sectional dependence and two lags of serial correlation, with country fixed effects and year dummies. The leave-one-country-out range is the smallest and largest coefficient obtained when each of the 28 countries is dropped in turn. Population-weighted estimates weight country-years by population.

Table B26: Large borrower-based reforms: first stage, reform-specific estimates and permutation inference

	Homeownership rate	Young mortgagors	Real household debt (%)	Real house prices (%)
All five reforms	0.570	-1.292	-0.152	-0.477
	(0.492)	(0.630)	(3.187)	(4.924)
Leaving out one reform				
Without Netherlands	0.044	-1.633	0.352	5.520
	(0.485)	(0.933)	(4.221)	(3.025)
Without UK	0.668	-1.731	-2.049	-2.743
	(0.562)	(0.690)	(3.071)	(5.241)
Without Ireland	0.377	-1.115	1.991	-3.172
	(0.565)	(0.683)	(2.959)	(4.769)
Without Sweden	0.753	-0.914	-0.314	-0.574
	(0.473)	(0.502)	(3.499)	(5.388)
Without Finland	0.780	-1.292	-0.347	0.170
	(0.458)	(0.695)	(3.504)	(5.538)
One reform at a time (other reforming countries excluded)				
Netherlands only	1.232	-0.938	-0.113	-9.719
	(0.402)	(0.554)	(3.425)	(2.925)
UK only	0.126	0.133	6.463	7.750
	(0.240)	(0.312)	(2.024)	(1.641)
Ireland only	1.372	-2.094	-11.277	12.205
	(0.187)	(0.310)	(1.663)	(1.271)
Sweden only	-0.935	-4.844	3.458	2.674
	(0.148)	(0.395)	(1.239)	(0.902)
Finland only	-1.135	-1.277	2.516	-5.393
	(0.148)	(0.395)	(1.239)	(0.902)
Control group restricted				
Controls without any borrower-based tightening	0.587	-0.680	0.428	1.461
	(0.569)	(0.775)	(3.180)	(5.053)
Permutation p-value, simple ATT (200 reassignments)	0.545	0.275		

Notes: Simple average treatment effects on the treated from the Callaway & Sant'Anna (2020) estimator with never-treated and not-yet-treated controls, for the aggregate homeownership rate and the share of young adult-only households owning with a mortgage (percentage points) and for the logs of real household debt and real house prices (times 100, so percent). The first row uses all five reforms. The next five drop one reforming country at a time. The following five use one reforming country against the never-treated countries only. The restricted control group excludes never-treated countries that tightened lending standards at some point in the sample. The permutation p-value reassigns the five treatment dates to five randomly chosen countries and reports the share of reassignments with an absolute effect at least as large as the estimated one. Standard errors clustered by country in parentheses.

Table B27: Large borrower-based policy shocks: staggered difference-in-differences estimates

	Pre-treatment	Post-treatment	Simple ATT	TWFE
Aggregate homeownership rate	0.784	0.570	0.570	-1.764
	(0.652)	(0.327)	(0.492)	(1.426)
Homeownership rate, adult-only HH under 65	1.781	0.302	0.638	-2.698
	(0.687)	(0.745)	(0.760)	(1.201)
Owner with mortgage, adult-only HH under 65	1.356	-1.312	-1.292	-4.413
	(0.729)	(0.533)	(0.630)	(1.393)
Renters, market rate	-0.331	-1.193	-1.207	0.372
	(0.676)	(0.458)	(0.580)	(1.237)
First stage (log times 100, percent)				
Real household debt	-0.823	-0.620	-0.152	-3.988
	(2.300)	(3.024)	(3.187)	(7.972)
Household debt to GDP	0.436	-2.498	-3.198	-8.893
	(3.679)	(5.449)	(6.978)	(12.035)
Real house prices	-6.429	-1.742	-0.477	3.499
	(3.319)	(4.033)	(4.924)	(6.128)

Notes: Treated countries and years: the Netherlands (2011), the UK (2014), Ireland (2015), Sweden (2016) and Finland (2016). The remaining 23 countries serve as never-treated controls, and not-yet-treated countries also serve as controls. The first three columns report the Callaway & Sant'Anna (2020) estimator: the average of the event-time coefficients over the three years before treatment, the average over the five years from treatment, and the simple average treatment effect on the treated. Pre-treatment coefficients are level differences relative to the year before treatment, so a positive pre-treatment average means that the outcome in treated countries was higher two and three years before treatment than in the reference year, relative to controls. The last column is the two-way fixed effects coefficient on a post-treatment indicator. The first four outcomes are in percentage points. The last three rows report the same statistics for the logs of real household debt, household debt relative to GDP and the real house price index, multiplied by 100, so the units are percent; Figure 9 plots the event-time coefficients for real house prices, real household debt, the aggregate homeownership rate and young mortgagors. Standard errors in parentheses are clustered by country.

C Online Appendix: Data

C.1 Macroeconomic data

I collect data from the national accounts for each country at the quarterly and annual level. The data is collected from Eurostat, unless otherwise indicated. From national accounts, I collect real GDP, final consumption expenditure, gross capital formation, total employment, and population. All variables are available at current prices or at purchasing power standard per capita. I also collect data on gross capital formation (divided into total fixed assets and total construction) in gross terms and as a share of GDP, and finally I collect the harmonised Indices of Consumer Prices (HICP) as a measure of inflation.

C.2 Credit, interest rates and house price data

I collect credit data from the BIS total credit statistics. The dataset includes credit data for a number of different countries and series, including credit to the non-financial sector and the government sector. The data is divided into two series on the lending side: Total credit and Bank credit. Total credit comprises lending from domestic banks, other domestic financial corporations, non-financial corporations and non-residents, whereas bank lending only includes credit from domestic banks to the private non-financial sector. On the borrowing side, total credit is broken down into the government sector and private non-financial sector, where the latter is further broken down into non-financial corporations and households (including non-profit institutions serving households).

The BIS total credit statistics do not cover ten of the 28 countries in the sample (Bulgaria, Croatia, Cyprus, Estonia, Latvia, Lithuania, Malta, Romania, Slovakia and Slovenia). For the specifications with control variables I therefore use Eurostat's private sector debt series (`tipspd20`), which measures the consolidated stock of loans and debt securities of non-financial corporations, households and non-profit institutions serving households as a share of GDP, and the corresponding household debt series (`tipspd22`). Both series cover all EU member states from 1995. Eurostat no longer publishes these series for the United Kingdom, for which I use

the BIS credit-to-GDP series for the private non-financial sector and for households. Annual growth rates of the BIS and Eurostat series have a correlation of 0.58 where both are available. Tables that use the BIS credit series are labeled as such.

House prices are measured with Eurostat's harmonized house price index (`prc_hpi_a`, all dwellings), deflated by the HICP and indexed to 2010. Eurostat does not publish an index for Greece, for which I use the BIS real residential property price index. For a few countries the Eurostat index starts after 2005 (Italy in 2010, Austria and Poland in 2008), and I extend it backward using the growth rate of the BIS index. Where both indices are available, their annual growth rates have a correlation of 0.99.

Short-term interest rates are measured with the 3-month money market rate from Eurostat (`irt_st_a`). For euro area members the euro area rate applies from the year of adoption, and national rates from Eurostat's historical series (`irt_h_mr3_a`) apply before adoption. Eurostat discontinued the national money market series for non-euro members after 2014, and for 2015 onward I use the BIS central bank policy rate for Croatia, Czechia, Hungary, Poland and Romania. Bulgaria, which operates a currency board, has no rate for these years. The interest rate enters the regressions as a first difference in percentage points. The BIS policy rate series, used in the robustness specifications, is missing for the same ten countries as the BIS credit data because it is only compiled for countries in the BIS credit statistics. The annual changes in the two rate series have a correlation of 0.97.[7]

C.3 Other sources of macroprudential policies

I collect data on macroprudential policies from the IMF's integrated Macroprudential Policy (iMaPP) Database, originally constructed by Alam *et al.* (2019). The database provides a dummy-type indicator for tightening and loosening of various macroprudential policy tools, along with a description of each policy action. The database combines information from five existing databases and the IMF's Macroprudential Policy Survey.[8] This dataset also contains

[7] For more information on the BIS data, see http://www.bis.org/statistics/pp.htm.
[8] More information about the database is available at https://www.elibrary-areaer.imf.org/Macroprudential/Pages/Home.aspx.

data on the average LTV ratio. I collapse policies to the annual level in order to match the frequency of the homeownership data.

I construct several indicators for policy actions. The ideal situation would be to have an indicator that captures both the intensity and the timing of any macroprudential policy change. However, since macroprudential policy instruments remain highly country-specific and often expand in scope over time, it is difficult to capture the intensity of a given policy action. I therefore construct policy indicators that capture the timing of policies, following most studies within the field. For instance, Akinci & Olmstead-Rumsey (2018) and Kuttner & Shim (2016) use an indicator that tracks whether a policy action was tightened, loosened or if there was no change. For country i at time t, the indicator $s^j_{i,t}$ denotes whether policy j was tightened, loosened or remained unchanged:

$$s^j_{i,t} = \begin{cases} 1, & \text{if policy instrument } j \text{ is tightened} \\ -1, & \text{if policy instrument } j \text{ is loosened} \\ 0 & \text{otherwise} \end{cases}$$

I use this indicator to define a net index $S^j_{i,k}$ as:

$$S^j_{i,t} = \sum_{k=0}^{t} s^j_{i,k} \tag{4}$$

This index represents the net cumulative number of policy actions taken by country i from time k until time t, where k is the earliest year where the homeownership data is available.